

Wonder World SCIENCE

A Textbook of Science for Preparatory Stage

5

As per New Education Policy (NEP) 2020 & NCF 2023



Wonder World **SCIENCE**

Class - 5

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Published by:

Silverline Publications

Printed at:

Capital Printers (Delhi)

ISBN 978-81-994864-2-3

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PREFACE

Wonder World SCIENCE is a series of science books for the students of classes 1 to 8. It has been designed in accordance with the National Education Policy (NEP 2020) and National Curriculum Framework (NCF 2023) visions. It ensures that each child gets the opportunity to learn multidisciplinary education through innovative means and methods. Our approach, as validated by the NCF, is interdisciplinary and incorporates the competencies goals. The NEP aims to support various learning paths that will include both formal and informal education. Making Indian students truly global citizens who are ready to step in future is the overarching objective. Keeping in mind the requirements, we have incorporated a perfect blend of Sustainable Development Goals (SDGs) encompass economic, social and environment dimensions.

The salient features of the series are:

- Each chapter begins with an Introduction that introduces the major topics it will cover.
- **Objective** gives a glance of topics that have been covered in the chapter.
- **Factopedia** provides some interesting facts related to the topic. These facts are accurate and have an extra spark of inspiration.
- **Key Words** help in explaining the important terms of the chapter.
- **Activity** is added inside the chapters to gain knowledge through playful activities.
- **Quick Recap** includes a brief summary of the chapter in the form of important points.
- **Abstract Analysis** includes objective as well as subjective questions as challenges for the students to retain and recall the concepts which they have studied.
- **Capability Analysis** is an extended learning which involves students actively by engaging them in real-world and personally meaningful projects. It covers scientific aptitude questions related to children's real-life situations which enriches learning as an interesting way followed by project or report working activities.
- **HOTS** questions help in developing the thinking skills of the students. It also covers value based questions which develop a strong relationship with surroundings.
- **Test Papers** are given at the end of each book to test the performance of the students.

We hope that the teachers and students will both be appreciative of our efforts. Any ideas for Improving the value of this series will be gratefully accepted. We eagerly await your comments and worthwhile recommendations for the series' continued improvement.

- Publisher

Special Features

When Should You Wash Your Hands?

After riding bus.

After playing

After Throwing garbage.

Before eating

Starter pages are given in each book to ignite a passion for environmental stewardship through captivating stories that inspire young minds to protect our planet.

Objectives : In this chapter gives a glance of topics that have been covered in the chapter

Objectives



• Types of Animals

• Birds and Insects

We see different types of animals around us. Animals are living things. They differ from each in many ways. Some animals are big. Some animals are small.

Observe the picture given below.

Critical Thinking



Put a tick (✓) near the birds, put a cross (×) near the insects and circle (○) the animals. Can you arrange the animals in the picture according to their size? (from biggest to smallest).

1. _____
2. _____
3. _____

An activity is provided as chapter starts to spark curiosity and critical thinking, setting the stage for scientific exploration.

Key Word help in explaining the important words of the chapter.

CEREALS AND PULSES

Some plants give us cereals like - wheat, rice, maize and pulses like - gram, peas, bean and...

Wheat

Millet

Rice

Gram

Pulses

Peas



Word
Cereals : grass-like plants which produce grains
Pulses : seeds of certain plants that can be eaten



Factopedia
Plants also give us tea, coffee and sugar.
Besides these, we get many other useful things from plants like wood, rubber, medicine, paper, etc.

Quick Recap includes a brief summary of the chapter in the form of important points.

Quick Recap

- * Plants give us fruits and vegetables.
- * Plants give us cereals and pulses.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. Plants give us:
 - a) fruits ☐
 - b) Vegetables ☐
 - c) B ☐
2. Which of the following are pulses?
 - a) Gram ☐
 - b) Peas ☐
 - c) B ☐

Abstract Analysis includes objective as well as subjective questions as challenges for the students to retain and recall the concepts which they have studied.

CAPABILITY ANALYSIS

A. Given below are pictures of some food items. Circle the food items with **blue** crayon that you take for breakfast and circle the food items with **red** crayon that you take for dinner.



Curd



Pulao



Rice



Vegetables



Milk



Egg

B. Rearrange the letters to get the names of some food we eat.

1. Geg _____
3. Dbrea _____
5. Cier _____

2. Lidi _____
4. Imlk _____
6. Team _____

Critical Thinking

HOTS

You did not eat breakfast yesterday and were feeling weak at school. Why do you think you were feeling weak?

HOTS questions help in developing the thinking skills of the students. It also covers value based questions which develop a strong relationship with surroundings.

Capability Analysis is an extended learning which involves students actively by engaging them in real-world and personally meaningful projects. It covers scientific aptitude questions related to children's real-life situations which enriches learning is an interesting way followed with project or report working activities.

Test Paper 2

(Based on Chapter 1-6)

A. Tick (✓) the correct option.

1. We eat _____ at night.
a) dinner ☐ b) lunch ☐ c) breakfast ☐
2. We wear gumboots in _____.
a) summer ☐ b) winter ☐ c) rainy season ☐
3. Which one causes damage?
a) Air ☐ b) Wind ☐ c) Storm ☐
4. What is the main source of water?
a) River ☐ b) Rain ☐ c) Lake ☐
5. We see this in the sky after it rains.
a) Clouds ☐ b) Rainbow ☐ c) White sky ☐

st ☐ c) south ☐

Column B

- a) zebra crossing.
- b) Swimming
- c) open and safe place
- d) dustbin
- e) before getting in or out

Test Papers are given at the end of each book to test the performance of the students.



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From Seeds to Harvest

Objectives



- *How Plants Grow from Seeds*
- *Agriculture*
- *Seed Dispersal*
- *Vegetative Propagation*
- *Seeds*
- *Crop Protection*

Plants play an important role in our surroundings. They produce their own food and serve as a primary food source for humans and animals. We rely on plants for fruits, cereals, and vegetables, which support us. Additionally, they provide herbs and medicines, clean the air, and supply oxygen. Plants also yield fibres for clothing and useful materials such as tea, coffee, jute, gum, and wood.

Create a Home Garden with Seedless Plants

Hand-On Learning

You can grow an indoor garden using top parts of vegetables like potatoes, onions, turnips, and carrots. Place them in a dish, ensuring they have some space between them. Pour enough water to cover the vegetables halfway, adding water every two to three days to maintain the level. After a week or two, small seedlings will sprout from the top parts of vegetables. Move them to pots for a beautiful indoor garden.



HOW DO PLANTS GROW FROM SEEDS?

A single plant can produce many seeds, but not all of them grow into new plants. Several factors can stop seed growth, including:

Lack of enough air, water, or warmth. Damage caused by wind and rain. Being eaten by birds, rats, or other animals.

When seeds receive the right amount of water, air, and sunlight, they begin to grow into new plants. This process is called sprouting.

SEEDS

While most plants **germinate** from seeds, some can grow from roots, stems, or leaves.

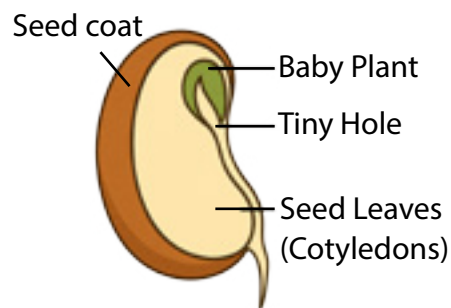
PARTS OF A SEED

Seed Coat

A thick outer layer that protects the seed. It has a tiny hole that allows the seed to absorb water.

Seed Leaves (Cotyledons)

These store food for the baby plant. The seedling grows between the cotyledons, which provide food and protection. Some seeds have two cotyledons, such as gram and bean, while others, like rice and wheat, have only one.



germinate : to grow

STAGES OF GERMINATION

A single plant can produce a large number of seeds. But not all seeds grow into new plants. The reasons are as follows:

- Some seeds do not get sufficient air, water or heat.
- Some seeds may be destroyed by wind and rain.
- Some seeds are eaten by birds, rats, and other animals.

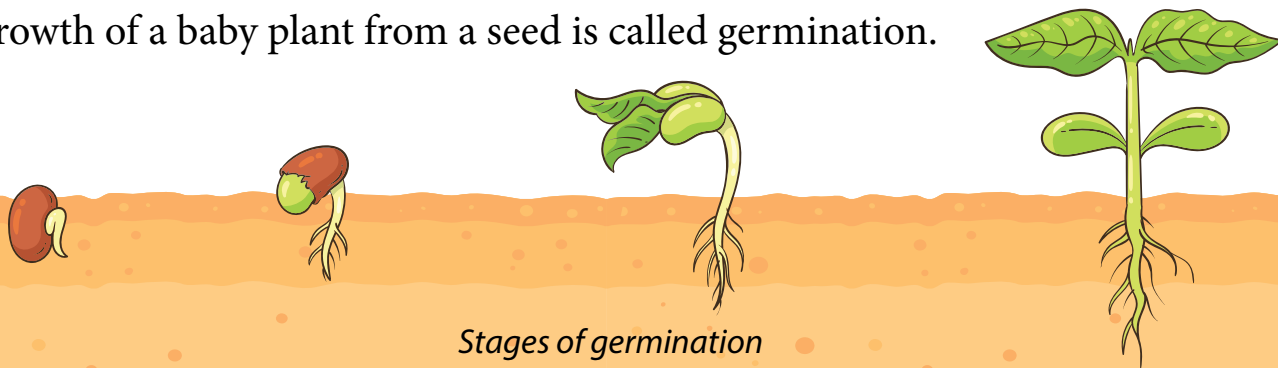
So, they are not able to survive. The seeds grow to become new plants when they get water, air, and sunlight in the proper amount.

The growth of a baby plant from a seed is called germination.



Did You Know

The largest seed in the plant kingdom is Coco de Mer, which weighs 27.22 kg.



Stages of germination

Activity

Soak some gram seeds and place them in different conditions:

- On a windowsill, exposed to sunlight and air.
- In a beaker filled with water.
- Inside a refrigerator.
- In moist sawdust, watered daily.

Observe the changes and discuss your findings with your teacher.

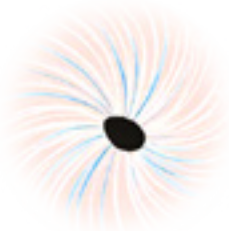


SEED DISPERSAL

Since plants cannot move, they rely on external forces to scatter their seeds, a process known as seed dispersal. Different methods include:

Dispersal by Wind

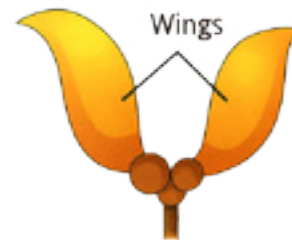
Some seeds are lightweight and have wings or hairs that help them float in the air. Examples:



Cotton (seed)



Madar (seed)



Hiptage

Dispersal by Water

Some seeds are lightweight and get dispersed by water. For example, the lotus fruit contains a spongy part, and the coconut has a thick fibrous coat, both of which help them stay afloat. When these fruits fall into the water, they float and are carried away by the stream to distant places.



Coconut (Fruit)



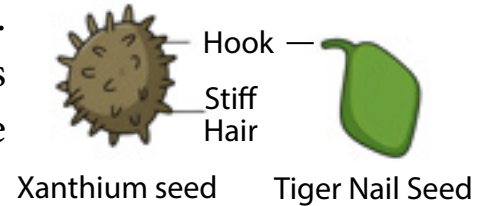
Lotus Outer Coat



Lotus

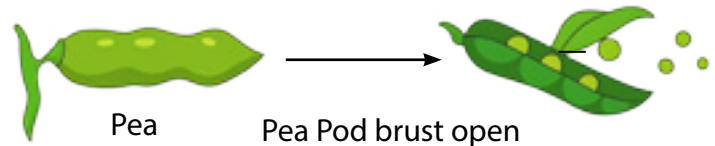
Dispersal by Animals

Some seeds have hooks, stiff hairs, or spines that stick to animal fur or our clothes. Examples are Xanthium, Spear Grass, and Tiger Nail Seed. Also, animals help in seed dispersal when they eat fruits like apple, mango, or guava and drop the seeds somewhere else. This helps new plants grow in different places.



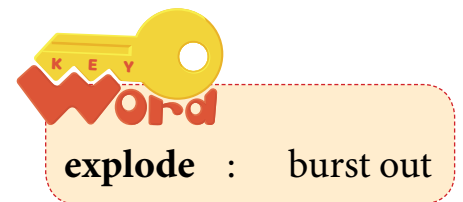
Dispersal by Explosion

Some fruits burst open when they become dry and throw their seeds in different directions. This helps the seeds spread far from the parent plant. Examples of such fruits are pea pods and balsam.



VEGETATIVE PROPAGATION

Certain plants can reproduce from their roots, stems, or leaves. This method is called **vegetative propagation**.

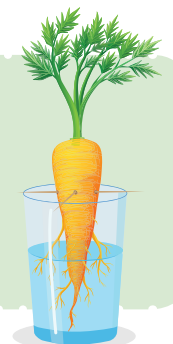


Plants from Roots

Some plants can grow new plants from their roots. This means they can sprout without seeds. Examples of such plants are carrot, radish, sweet potato, and turnip.

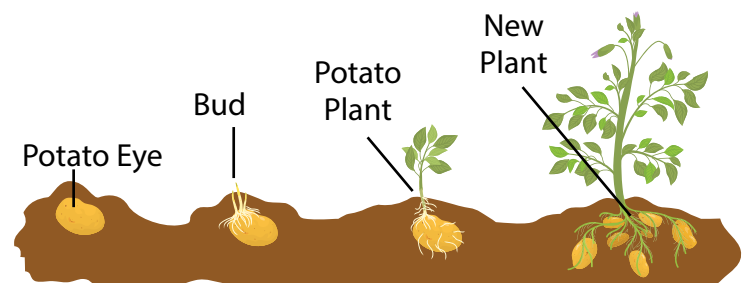
Activity

1. Take a carrot and put it in a beaker filled with water.
2. Support it with toothpicks so only the bottom is submerged.
3. Keep the beaker in an open area with air and sunlight.
4. Observe the new plant growth after a few days.



Plants from Underground Stems

Some plants like ginger, onion, and potato grow from underground stems. Potatoes and ginger have small spots called “eyes” or buds. These buds grow into new plants when planted in the soil.



A part of potato that has an eye can grow into a new plant.

Plants from Stems

Some plants like rose, hibiscus, sugarcane, and cotton can grow when a part of their stem is cut and planted in soil.



Stem Cutting of Sugar-cane



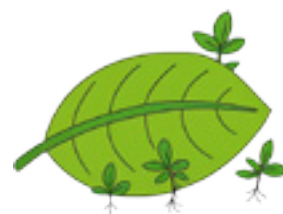
Sown in Ground



New Plant of a Sugar-cane

Plants from Leaves

Plants like Bryophyllum grow new plants from buds on their leaf edges when placed in moist soil.



Bryophyllum Leaf

Plants from Spores

Some plants, like ferns, mushrooms, and moss, do not make seeds like other plants. Instead, they grow from tiny powder-like parts called **spores**. These spores can travel through air or water and grow into new plants when they land in the right place.



spore : small structure like seed



Fern



Mushroom



Moss

AGRICULTURE

Agriculture means growing crops and raising animals to get food, clothes, and other useful things. It is very important because it gives us rice, wheat, fruits, vegetables, milk, eggs, meat, cotton, and many other things we use every day.

Most of the food we eat and the clothes we wear come from plants and animals. Farmers work hard in fields to grow crops and take care of animals. This whole process is called farming or agriculture.

STAGES OF AGRICULTURE



1. Ploughing

Loosening the soil.



2. Manuring

Enriching the soil with nutrients.



3. Sowing Seeds

Beginning crop production.



4. Irrigation

Providing water to plants.



5. Spraying Pesticides

Protecting crops from pests.



6. Harvesting

Cutting and collecting mature crops.



Quick Reflection

1. What is ploughing?
2. Define crop.

When plants of the same kind are grown at one place on a large scale, it is called a **crop**. For example, millet, wheat, etc.

Kharif and Rabi Crops

Farmers grow different types of crops depending on the season. In India, there are two main crop seasons—Kharif and Rabi. Kharif crops are grown during the rainy season, from June to October. These crops need a lot of water to grow. Examples of Kharif crops are rice, maize (corn), and cotton. On the other hand, Rabi crops are grown in



Kharif crops

the winter season, from November to April. These crops grow well in cool weather and need less water compared to Kharif crops. Common Rabi crops include wheat, gram (chana), and peas. Farmers choose which crops to grow based on the weather, rainfall, and temperature of each season.



Rabi crops

CROP PROTECTION



Crops need protection from birds, insects, rats, and other pests that can harm them. Farmers use different methods to keep the crops safe. They spray pesticides and **insecticides** on the plants to keep away harmful insects. To protect crops from rats and mice, they use rodent-repelling chemicals. After harvesting, farmers use proper storage methods to keep the grains dry and safe from insects and moisture. This helps

keep the food fresh and safe for eating.



insecticides : chemical to kill insects that eat plants

Quick Recap

- ✧ Most plants grow from seeds.
- ✧ Seeds are dispersed by wind, water, animals, or explosion.
- ✧ Germination is the process of a seed growing into a plant.
- ✧ Some plants reproduce through vegetative propagation.
- ✧ Kharif crops grow in the rainy season, while rabi crops grow in winter.
- ✧ Agriculture involves growing plants and raising livestock for food.
- ✧ Insecticides help protect crops from pests.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

- Plants can grow from:
a) Seeds ☐ b) Leaves ☐ c) Both a & b ☐
- The outer protective layer of a seed is called:
a) Cotyledons ☐ b) Seed coat ☐ c) Germination ☐
- Which seed is NOT dispersed by water?
a) Coconut ☐ b) Lotus ☐ c) Cotton ☐
- Cotton seeds are mainly dispersed by:
a) Air ☐ b) Water ☐ c) Explosion ☐
- Which is a Kharif crop?
a) Wheat ☐ b) Legumes ☐ c) Rice ☐

B. Fill in the blanks with the appropriate words from the box.

Seeds crop Bryophyllum dispersal Wheat

- _____ germinate and grow into new plants.
- The scattering of seeds away from their mother plant is called _____
- _____ plant grow from the leaf buds.
- When plants of same kind are grown at one place on a large scale, it is called _____.
- _____ is a rabi crop.

C. Match the following.

Column A

- Seed dispersal by animal
- Rabi crop
- Seed dispersed by explosion
- Baby plant
- Seed dispersed by water

Column B

- Seedling
- Coconut
- Poppy
- Pea
- Xanthium

D. Write T for true and F for false statements.

- Cotyledon provides food to seedling. ☐
- Seed coat is a part of stem of the plant. ☐
- The growth of a seed into seedling is called bursting. ☐

4. Seeds of pea plant disperse by explosion.
5. Maize crop is grown in November.



E. Answer the following questions.

1. Name the three things that seed require to germinate.
2. Describe the process of germination.
3. What is seed dispersal ? What are various agents of seed dispersal?
4. Which parts of the plants are used to grow new plants?
5. What are kharif and rabi crop? Give examples.

CAPABILITY ANALYSIS

- A** Collect information on different kinds of crops grown in India. Write a report.

Knowledge of India

- B.** Take a potato. Support it in a bottle filled with water using some toothpicks. Keep the lower part of the potato immersed in water. What do you observe after a few days?

Experiential Learning

- C.** Visit a farm or farmhouse if you can, and talk to the farmer or gardener there. You can inquire about the various steps they follow to grow crops or other plants.

Experiential Learning

D. Innovation in the Agriculture Industry

To support a growing population, many farmers are turning to new technologies. Smart farming uses information and communication technologies such as the Internet of Things (IoT), robotics, and augmented and virtual reality (AR/VR) to improve both the quantity and quality of products while reducing human labour.



Multidisciplinary Learning

Smart farming offers farmers many high-tech tools that provide valuable data to help them make better decisions. Explore the Internet to find out more about some of the current technologies. Make a PowerPoint presentation on “The Future of Farming.”

Digital Resources

HOTS

Cotton seeds are dispersed by air. Why?

Critical Thinking

Food and Health

Objectives

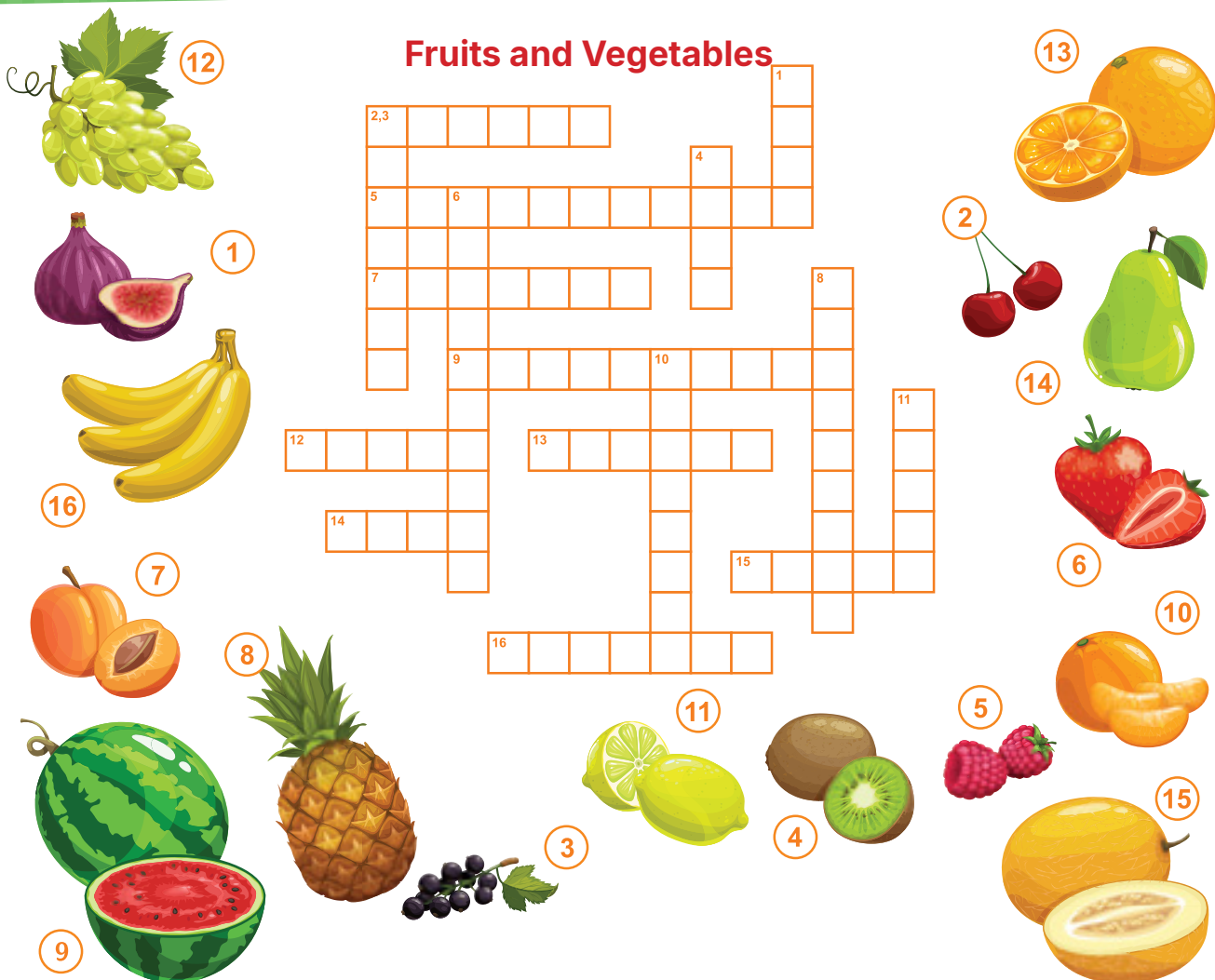


- *Balanced Diet*
- *Diseases and Their Symptoms*
- *Junk Food*
- *Deficiency Diseases*

Food is an essential requirement for our survival. It supports our growth, provides energy for daily activities, and keeps us healthy. A variety of foods is necessary to maintain good health. Some food items come from plants, while others are obtained from animals.

Solve the Crossword Puzzle of fruits and vegetables.

Problem Solving



HEALTH

Health refers to a state of complete physical and mental well-being. To stay healthy, we should live in a clean environment, breathe fresh air, drink safe water, and follow a balanced diet.

BALANCED DIET

The daily food intake of a person is called a **diet**. A balanced diet contains all the essential nutrients—carbohydrates, fats, proteins, vitamins, and minerals—in proper amounts.



- Carbohydrates and fats supply energy to the body.



Carbohydrates and fats

- Proteins support muscle development and body growth.



Proteins

- Vitamins and minerals help the body fight diseases and keep us healthy. Fruits and vegetables are good sources of these nutrients.



- Water and roughage are also essential. We should drink 8–10 glasses of water daily and consume fiber-rich foods to aid digestion.



Vitamins and minerals

- Milk is considered a complete food. We should drink two glasses of milk daily and include dairy products like curd, cheese, and paneer in our diet.



Milk products

Did You Know

Soyabean contains the highest amount of protein among plant-based foods.



adequate : proper

JUNK FOOD

Junk food is unhealthy as it is high in calories, fats, and sugar but low in fibre, proteins, vitamins, and minerals.



Burger



Soft Drink



Chips



Noodles



Pasta



Cake



Pizza



Ice Cream

Harmful Effects of Junk Food

1. It creates a temporary feeling of fullness but provides little nutrition, leading to tiredness.
2. The high fat content can result in unhealthy weight gain.
3. Excessive salt and sugar in junk food can increase blood pressure and sugar levels.
4. It is difficult to digest as it is highly processed.
5. It contributes to dental problems like tooth decay.

To maintain good health, we should reduce junk food intake.

Activity

Activity: Sprouting Green moong Dal

Objective: To prepare healthy sprouts.

Materials Needed: Green Moong Dal, Water, Bowl and Muslin Cloth.

Method: Rinse moong dal and soak it overnight in water.

Drain the water the next morning and transfer the soaked dal to a muslin cloth.

Tie the cloth securely and keep it aside for 8–10 hours.

After the required time, you will notice sprouts emerging. These sprouts are highly nutritious and beneficial for health.



DISEASES AND THEIR SYMPTOMS

A disease is a condition that affects the body and makes us feel unwell. Many diseases are caused by poor lifestyle choices like unhealthy eating and lack of exercise. Junk food, too much sugar or fat, and skipping meals can lead to problems like obesity, diabetes, and heart disease. To stay healthy, we should eat a balanced diet, exercise often, and maintain good habits.



Obesity

Obesity is excessive weight gain caused by overeating, lack of exercise, or an unhealthy lifestyle.



An Obese Person

Symptoms:

- Digestive issues
- Fatigue
- Body pain
- Dry and scaly skin

Diabetes

Diabetes occurs when the body cannot regulate blood sugar levels properly. It is common in adults but is increasingly affecting children due to poor dietary habits.

SYMPTOMS IN CHILDREN

- Fatigue
- Sudden weight loss or gain
- Frequent infections
- Increased thirst and urination

High Blood Pressure

High blood pressure occurs when the heart has to pump blood at high pressure, putting a strain on the heart and blood vessels. It is often caused by high salt intake and junk food consumption.



SYMPTOMS

- Dizziness
- Nausea
- Blurred vision
- Nosebleeds
- Headache

Anaemia

Anaemia is a condition where the body has low red blood cell (RBC) count due to iron deficiency.

Symptoms in Children:

- Weakness and fatigue
- Shortness of breath
- Loss of consciousness
- Pale appearance



Anaemia child

DEFICIENCY DISEASES

When the body lacks essential nutrients, it can lead to various deficiency diseases.



deficiency : lack of something

Marasmus

A disease affecting children under one year due to severe deficiency of proteins and carbohydrates.

SYMPTOMS

- Extreme weight loss
- Dry and wrinkled skin
- Stunted physical and mental growth
- Weakness



Marasmus

Prevention: Include protein-rich foods like pulses, soybeans, eggs, nuts, and fish in the diet.

Kwashiorkor

A disease caused by protein deficiency in children aged 1–3 years.

SYMPTOMS

- Thin, weak legs
- Bloated stomach
- Stunted growth and mental retardation
- Loss of appetite

Prevention: Eat protein-rich foods like pulses, nuts, eggs, and fish.



Kwashiorkor



Did You Know

Kwashiorkor and Marasmus are together termed as PEM, i.e., Protein Energy Malnutrition.

Night Blindness

Night blindness is a condition where a person has difficulty seeing in low light or darkness. It is often caused by a deficiency of Vitamin A, a nutrient essential for healthy vision, especially in dim lighting. Without enough Vitamin A, the eyes can't properly adjust to darkness, making it hard to see at night or in poorly lit places.



Prevention: Consume Vitamin A-rich foods like carrots, spinach, and papaya.



Beri-Beri

Beriberi is a disease caused by a deficiency of Vitamin B, especially Vitamin B1 (thiamine). It affects the nervous system and can lead to nerve damage, muscle weakness, and weight loss. To prevent beriberi, eat foods rich in Vitamin B such as whole cereals, pulses, nuts, milk, and green vegetables. A balanced diet helps keep the nervous system healthy.



Scurvy

Scurvy is a disease caused by a lack of Vitamin C, which is important for healthy skin, gums, and bones. Symptoms include bleeding gums, loose teeth, muscle pain, fatigue, and weakness.

It can be prevented by eating Vitamin C-rich foods like oranges, lemons, tomatoes, strawberries, and leafy green vegetables.



Rickets

Rickets is a disease caused by a deficiency of Vitamin D, which is essential for healthy bones and teeth. Lack of Vitamin D leads to weak and deformed bones, especially in children. Common symptoms include bow-shaped legs and soft deformed ribs. To prevent rickets, eat Vitamin D-rich foods like egg yolk, milk, and fish, and get regular sunlight exposure, which helps the body produce Vitamin D naturally.



Rickets



Goitre

Goitre

Goitre is a disease caused by iodine deficiency, which leads to the enlargement of the thyroid gland, seen as a swelling in the neck. It can be prevented by using iodized salt in daily cooking and eating iodine-rich foods like leafy vegetables, dairy products, and seafood.

Quick Recap

- * A balanced diet contains all nutrients in the right amounts.
- * The main food components are carbohydrates, fats, proteins, vitamins, minerals, water, and roughage.
- * Junk food is harmful and should be consumed in limited quantities.
- * Lifestyle diseases include obesity, diabetes, high blood pressure, and anaemia.
- * Deficiency diseases occur due to a lack of certain nutrients.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. Which of the following represents a healthy diet?
a) Sprouts ☐ b) Milk ☐ c) Both a) & b) ☐
2. _____ aid in muscle development and promote body growth.
a) Carbohydrates ☐ b) Burger ☐ c) Proteins ☐
3. A deficiency of which vitamin causes night blindness?
a) Vitamin A ☐ b) Vitamin B ☐ c) Vitamin D ☐

4. In which disease is iodine in salt essential?

a) Anaemia

☐

b) Goitre

☐

c) Beri-beri

☐

B. Fill in the blanks with the appropriate words from the box.

sugar

ribs

bones

Vitamin D

Kwashiorkor

Vitamin C

1. In diabetes, there is an increase in _____ levels.

2. _____ is caused due to protein deficiency in children between 1-3 years.

3. Scurvy is a deficiency disorder resulting from a lack of _____.

4. Rickets leads to deformities in _____ and _____.

5. Our body produces _____ when exposed to sunlight.

C. Match the following.

Column A

1. Anaemia

2. Beri-Beri

3. Scurvy

4. Rickets

5. Goitre

6. Night Blindness

Column B

a) Vitamin D

b) Vitamin C

c) Vitamin B

d) Vitamin A

e) Iron

f) Iodine

D. Identify and Describe These Diseases

1.



Cause: _____

Symptoms: _____

Prevention: _____

2.



Cause: _____

Symptoms: _____

Prevention: _____

3.



Cause: _____

Symptoms: _____

Prevention: _____

4.



Cause: _____

Symptoms: _____

Prevention: _____

E. Answer the Following Questions

1. What is a balanced diet? List its components.
2. Define junk food and explain its negative effects.
3. What are the symptoms of diabetes?
4. Define anaemia and mention its symptoms.
5. What is Marasmus? How can it be treated?
6. What is PEM?

CAPABILITY ANALYSIS

A. Scrapbook Activity

Multidisciplinary Learning

Document the symptoms and preventive measures of the following conditions in your scrapbook: Blood Sugar High, Blood Pressure and Rickets

B. Poster Making

Create an informative poster highlighting the benefits of iodized salt.

Art Integration



C. Crossword Puzzle Activity

Problem Solving

Find and circle the names of diseases hidden in the crossword puzzle.

R	O	T	M	D	N	P	B	E	A
I	Q	G	O	I	T	R	E	A	T
C	O	O	B	A	M	S	R	N	S
K	I	T	E	B	A	T	I	A	C
E	A	T	S	E	R	U	B	E	U
T	E	N	T	T	A	V	E	M	R
S	E	A	T	E	S	W	R	I	V
Z	Y	X	Y	S	M	A	I	A	Y
A	B	C	D	J	U	N	P	O	T
E	F	G	H	I	S	U	G	A	R

D. Research Activity

(Charaka was an esteemed ancient Indian physician and scholar, regarded as the father of Ayurveda. He made significant contributions to Ayurvedic medicine and authored the Charaka Samhita, a comprehensive text detailing numerous diseases and their treatments.) Use the internet to gather information about Charaka's life, his contributions to Ayurveda, and the significance of the Charaka Samhita. Compile your findings into a Word document.



Digital Resources

E. COVID-19 Awareness

Corona virus is a newly discovered virus that causes an illness called COVID-19. Symptoms may include fever, cough, sore throat, and difficulty in breathing.

Discuss safety measures and hygiene protocols for protection against such infections in your class.

Communication Skill

HOTS

Why is it essential for our body to receive all nutrients to maintain good health?

Critical Thinking

Safety and First Aid

Objectives



- Road Safety

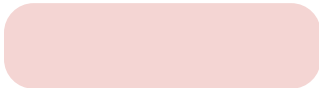
- First Aid

Whether you are working, studying, or playing, you are constantly moving and handling different things, which can sometimes lead to injuries. Accidents can occur due to negligence—either yours or someone else's. Both children and adults can get hurt at home, in school, on the playground, or while traveling on the road.

What do the following road signs indicate?

Critical Thinking

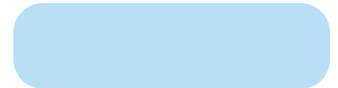
1.



2.



3.



4.



5.



6.



7.



8.



9.



ROAD SAFETY

To prevent accidents, everyone must follow basic road safety rules:



Always use a zebra crossing while crossing the road.



Never board or exit a moving bus.



Avoid leaning out of a moving vehicle.



Do not distract the driver while the vehicle is in motion.



Refrain from jumping on desks.



Always stand in a queue at bus stops.



Do not use blades for sharpening pencils.



Keep toys off the floor to prevent tripping hazards.



Do not use blades for sharpening pencils.



Do not run on staircases.



Never swim alone.



Do not touch electric wires.

FIRST AID

First aid refers to the immediate medical assistance given to an injured person before professional medical care is available. Quick and timely help can save lives. However, first aid is mainly for minor injuries such as small wounds, nosebleeds, fatigue, dizziness, and minor burns. It is essential to have basic first-aid knowledge to assist those in need.

General First Aid Guidelines:

- Always wash hands before administering first aid to prevent infections.

Sprains

A **sprain** occurs when a joint, such as the ankle, gets twisted, causing swelling and pain. The affected joint should be kept still until healed.

First Aid for Sprains:

- Apply an ointment to the sprained area 2-3 times daily.
- Wrap a crepe bandage around the joint after applying the ointment.



Fractures

A fracture is a break or crack in a bone, causing intense pain and swelling. First Aid for **Fractures**:

- Keep the injured person still.
- Help the person remain calm and comfortable.
- Use a splint to support the broken bone.
- If the arm is fractured, support it with a sling to prevent movement.



Burns



Burns can be caused by hot objects, steam, fire, electric shocks, or chemicals like acids. Minor burns result in reddened skin.

First Aid for Burns: Rinse the burnt area with cold water. Apply Burnol and cover with a clean cloth to prevent infection.



Did You Know

A sling is a piece of cloth tied around the neck to support an injured arm.

For severe burns with blisters, never prick them—apply a wet compress with baking soda solution. In case of chemical burns, wash the affected area with plenty of water and seek medical attention immediately.



infection : diseases that can be passed from person to another

Minor Cuts and Wounds

- Wash the wound with soap and water, then clean with antiseptic-soaked cotton (Dettol or Savlon).
- If bleeding, applying ice helps reduce blood flow.



Deep Cuts

- Clean the wound with antiseptic-dipped cotton.
- To slow bleeding, tie a bandage tightly above the wound.
- Applying firm pressure with a cotton pad can also help stop bleeding.
- A **tourniquet** (a tightly tied bandage) may be used in severe cases.
- Seek medical attention immediately.



Did You Know

Tetanus is a disease caused by bacteria entering through dirty or rusty wounds. A tetanus injection, given within 24 hours, prevents infection.

Nose Bleeding

Nosebleeds are common in children, especially during summer. First Aid for Nosebleeds:

- Make the person sit comfortably and tilt their head back.
- Place an ice pack on the nose to slow bleeding.
- Encourage breathing through the mouth—do not let them blow their nose.
- If bleeding persists, seek medical help.



Fire Safety

Fires can be extremely dangerous, spreading rapidly and causing significant damage. Understanding fire hazards and preventive measures is essential.

Common Causes of Fire:

- Careless use of matchsticks, candles, and firecrackers.
- Gas leaks in kitchens.
- Electrical malfunctions.

Fire Prevention Tips:

- Handle matchsticks, candles, and firecrackers with care.
- Store petrol and kerosene in sealed containers away from flames.
- Avoid wearing synthetic clothes while cooking.
- Always turn off gas cylinders after use.
- Regularly check electrical wiring.

Ways to Extinguish Fire:

- Fire extinguishers (containing CO₂ or foam) smother fires by cutting off oxygen.
- Petrol fires can be put out with sand or mud.
- Electrical fires should be tackled by switching off the mains and using sand or mud, never water.
- For large fires, call the fire brigade by dialling 101 or 112.



Poisoning

Poisoning can be life-threatening, so toxic substances must be kept out of children's reach. Preventive Measures:

- Always check expiry dates on medicines before consuming them.
- Wash fruits and vegetables thoroughly to remove pesticide residue.
- Items like nail polish, naphthalene balls, paint, and detergents are toxic.

First Aid for Poisoning:

- Induce vomiting if the poison is ingested.
- Seek medical assistance immediately.

Animal Bites

Animal bites can be dangerous as their saliva may contain harmful viruses like rabies. Animals that may spread rabies:

- Dogs, cats, monkeys, wolves, and jackals.

First Aid for Animal Bites:

- Wash the wound with soap and water.
- Apply antiseptic and bandage with sterile gauze.
- Seek medical attention immediately—rabies vaccines may be required.

Snake Bites:

- Apply a tight bandage above the bite to slow the spread of venom.
- Keep the person still to prevent the poison from circulating quickly.
- Seek medical help immediately.



saliva : liquid inside the mouth of animals and human

Quick Recap

- * Accidents often result from carelessness.
- * Following safety guidelines can prevent injuries.
- * Different injuries require different first aid measures.
- * Always treat wounds properly to prevent infections.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. What is the primary cause of accidents?

a) Carelessness

☐

b) Poor vision

☐

c) Thrills

☐

2. What is a common summer health issue for children?

a) Wounds

☐

b) Sprains

☐

c) Nosebleeds

☐

3. How can you put out an electrical fire?

a) With water

☐

b) Using a fire extinguisher

☐

c) With sand or mud

☐

4. Which of these is a good conductor of electricity?

a) Sand

☐

b) Mud

☐

c) Water

☐

B. Fill in the blanks with appropriate words from the box.

burnol

rabies

injured

viruses

bone

1. The first help provided to an _____ individual is known as first aid.

2. A break or crack in a _____ is referred to as a fracture.

3. To relieve minor burns, apply _____ on the affected area.

4. Animal saliva contains _____.

5. A doctor may administer _____ injections in case of an animal bite.

C. Write T for true and F for false statements.

1. Before assisting an injured person, you should wash your hands.

☐

2. If a nosebleed stops, a bandage called a tourniquet should be applied.

☐

3. A fractured part should be kept immobile to prevent further injury.

☐

4. Rabies, a severe disease, is caused by the saliva of animals such as dogs and cats.

☐

5. Naphthalene balls are toxic substances.

☐

D. Answer the following questions.

1. What is first aid?

2. How would you assist someone with a deep cut?

3. How would you help someone experiencing a nosebleed?

4. What is a fracture? What first aid should be provided?

5. How would you assist someone with a sprain?

CAPABILITY ANALYSIS

- A.** Create a poster highlighting the importance of being cautious to prevent accidents. Creativity
- B.** Visit a fire station, observe the operations of fire fighters, and document their equipment and responsibilities in a report. Experiential Learning
- C.** Assemble a customized first aid kit including alcohol wipes, hand sanitizer, cough drops, and an oximeter. Decorate it with first aid-related stickers.



Multidisciplinary Learning

- D.** Engage in a discussion with your teacher on the following topics: Life Skill
- Why electrical wires are coated with plastic.
 - What materials should not be used to extinguish a fire caused by a kerosene stove.
- E.** Research and list various types of fire extinguishers along with their specific uses. Scientific Temper
- F.** Develop a Fire Safety Guide outlining fire prevention measures and emergency response strategies. Vocational Skill
- G.** Construct a model of a fire extinguisher using a red craft sheet and a clothes pin. Creativity
- H.** Collect old X-ray films from a doctor's clinic or an adult at home. Examine them to identify any images of fractured bones, then write a report based on your observations. Experiential Learning



HOTS

Why does Supriya's mother not allow her into the kitchen?

Life Skill

Rocks and Minerals

Objectives



- Rocks

- Minerals

- Types of Rocks

The Earth consists of various types of rocks and minerals. Rocks are natural materials found in the Earth's crust.

Match the following minerals and rocks with their names:

Critical Thinking



Quartz

Tungsten

Marble

Graphite

Diamond

Granite



Now, categorize the minerals and rocks:

Hard	Soft

ROCKS

Rocks are solid, rigid masses composed of mineral particles bonded together. Their composition and formation determine their types. The minerals within rocks make them unique in terms of origin, color, texture, and metallic content.

TYPES OF ROCKS

Rocks are classified into three categories based on their formation:

- Igneous Rocks
- Sedimentary Rocks
- Metamorphic Rocks



Igneous Rocks

Igneous rocks are the earliest type of rock, formed by the cooling and solidification of magma from deep within the Earth. When rocks melt at extremely high temperatures within the Earth's crust, they turn into magma. When magma reaches the Earth's surface, it forms lava. During volcanic eruptions, lava cools and solidifies to form igneous rocks. Rapid cooling produces pumice, obsidian, and basalt, while slow cooling results in granite.

Types of Igneous Rocks:

- **Granite:** The most common type of rock, very hard, and used for constructing floors, statues, and roads. It comes in black, grey, white, or pink.
- **Basalt:** Forms from rapidly cooled volcanic lava. It appears dark, shiny, and layered like glass with sharp edges. Used in making tools.

- **Pumice:** A soft, lightweight, and spongy rock formed from quickly cooled lava. Trapped gases create a porous texture. It is used in making stone utensils, teeth polishing, and as a skin scrubber.



Granite



Basalt



Pumice

Sedimentary Rocks

Sedimentary rocks originate from pebbles, sand, mud, or clay that accumulate in oceans. Water currents and floating ice, transport these sediments to shore, forming layers. These layers compress over time and cement together with minerals, creating sedimentary rocks.

Types of Sedimentary Rocks:

Sandstone: Created from compacted sand. Soft and found in yellow, brown, pink, or red hues.

Limestone: Composed mainly of calcium carbonate, often derived from shells, corals, and bones. Used in making bricks, glass, quicklime, chalk, and cement.

Shale: Formed from clay and mud in riverbeds, it is gray and splits into flat layers. Used for making tiles and bricks, as well as in cement production.

Conglomerate: Composed of solidified gravel, stones, and pebbles. Hard and irregular in shape, it was historically used as a millstone for grinding corn.

Dolomite: Primarily composed of magnesium and used in cement production and magnesium extraction.

? Did You Know

Pumice is the only rock that floats on water.

Goldsmiths use basalt to test the purity of gold.



Sandstone



Limestone



Shale



Conglomerate



Dolomite

Quick Reflection



1. How many types of rocks exist?
2. Which is the oldest type of rock?

Metamorphic Rocks

The term “metamorphic” means “changed form.” These rocks develop when existing igneous or sedimentary rocks undergo transformation due to extreme heat or pressure. The minerals in the rocks restructure, forming tough, durable metamorphic rocks.

Examples of Metamorphic Rocks:

- **Marble:** A hardened, crystalline form of limestone. It is smooth, white, and used for flooring and statues. The TajMahal is built from white marble.
- **Mica:** A shiny, transparent mineral found in thin layers, used in electrical appliances and spacecraft due to its resistance to heat and electricity.
- **Quartzite:** Extremely hard and unsuitable for construction. It originates from sandstone and is used in statues and watches.
- **Slate:** A hardened form of shale, smooth, black, and layered. Easily split into thin sheets, it is used for blackboards and slates.
- **Gneiss:** A metamorphic version of granite with light and dark bands. Polished gneiss is used for flooring tiles.



Marble



Mica



Quartzite



Slate



Gneiss

MINERALS

Minerals are natural underground resources, and all rocks contain them. Minerals that have high metallic content and can be mined profitably are called ores. Examples include: Hematite (iron ore), Bauxite (aluminium ore), Pyrite (copper ore) Calamine (zinc ore)



Aluminium utensils



Copper wires



Gold jewellery

Coal

Coal is a crucial fossil fuel, formed from plant and animal remains buried for millions of years. Over time, heat and pressure turn these remains into coal, which is rich in carbon.

Uses of Coal:

- Used as fuel for cooking.
- Powers steam engines, factories, and furnaces.
- Generates electricity in power plants.
- Used in metal extraction.

India's major coal mines are located in Dhanbad, Raniganj, Neyveli, Singaneri, and Suhagpur.

Types of Coal: Peat, Lignite, Bituminous and Anthracite.



Uses of Coal:

Petroleum

The term “petroleum” is derived from Latin words: “Petra” (rock) and “Oleum” (oil). It is a mineral oil found in rocks, formed from ancient plant and animal remains. Over millions of years, these remains transformed into oil and gas. Petroleum is a primary energy source, extracted through fractional distillation to produce fuels such as petrol, diesel, kerosene, and fuel oil.

Petroleum is often referred to as “liquid gold.”



Did You Know

Fractional distillation separates liquids that dissolve in each other, known as miscible liquids.

Quick Recap

- * Rocks are natural materials found in the Earth’s crust.
- * They mainly consist of silicate and minerals.
- * Rocks contain one or more minerals.
- * Igneous rock is the oldest type of rock.
- * Minerals that contain metal deposits are called ores.
- * Coal and petroleum are fossil fuels..

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. Which of the following is an igneous rock?

- | | | | |
|-----------------|--------------------------|-----------|--------------------------|
| a) Pumice | ☐ | b) Basalt | ☐ |
| c) Both a) & b) | ☐ | | |

2. Which rock was historically used for grinding corn?

- | | | | |
|--------------|--------------------------|-----------------|--------------------------|
| a) Limestone | ☐ | b) Conglomerate | ☐ |
| c) Shale | ☐ | | |

3. Which is a metamorphic form of granite?

a) Gneiss

☐

b) Basalt

☐

c) Mica

☐

4. Coal is rich in _____ and serves as a fuel source.

a) Carbon

☐

b) Nitrogen

☐

c) Hydrogen

☐

B. Fill in the blanks with the appropriate words from the box.

fossil

white marble

Diamond

Petroleum

igneous

1. Rocks formed due to cooling lava are known as _____ rocks.

2. Taj Mahal is made up of _____

3. _____ is the hardest substance found on the Earth.

4. Coal is a _____ fuel.

5. _____ is also known as liquid gold.

C Write T for true and F for false statements.

1. Marble is an impure form of limestone.

☐

2. Coal can be used to generate electricity.

☐

3. Diesel is obtained from petroleum.

☐

4. All minerals are ores.

☐

5. Paraffin wax is obtained from coal.

☐

D. Answer the following questions.

1. What are sedimentary rocks? Give two examples.

2. Which rocks are formed by cooling of lava ?

3. Name and explain any two metamorphic rocks.

4. How is petroleum formed? What are its derivatives?

5. Write short note on rocks used in construction of buildings.

CAPABILITY ANALYSIS

A. Visit a jewelry store with your mother to observe and identify different precious stones.

Experiential Learning

B. Discuss the significance of coal and petroleum.

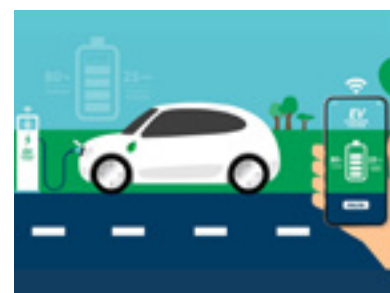
Communication Skills

C. Research and create a state-wise list of rocks and minerals found in India.

Knowledge of India

D. Electric vehicle (Evs) are becoming increasingly popular as more people are looking for sustainable and eco-friendly transportation solution.

Digital Resources



Prepare a Power Point presentation on pollution and modern fuel substitutes.

Environment and Health

E. Research historic monuments such as the Parliament House, Qutub Minar, and Amber Fort, noting the stones used in their construction.

Knowledge of India



HOTS

Which type of rock is used for flooring in homes?

Critical Thinking

Adaptation in Animals

Objectives

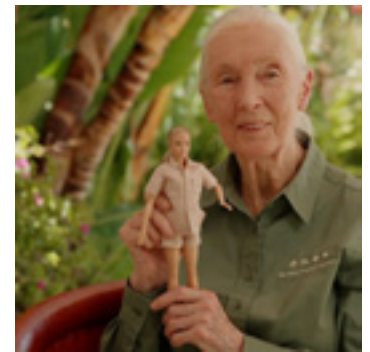


- *Habitat*
- *Movement in Animals*
- *Migration*
- *Respiration in Animals*
- *Adaptations of Animals in Different Habitats*
- *Eating Habits of Animals*

You can observe a variety of animals around you—some live on land, some in water, while others are found in the air.

Jane Goodall is widely recognized for her pioneering research on chimpanzees. If her career inspires you and you wish to pursue a similar path in primatology or animal behaviour research, here are three steps you could take:

1. Gain knowledge in fields like zoology, biology, or environmental science.
2. Participate in wildlife conservation projects or research programs.
3. Work with organizations dedicated to animal behaviour studies and conservation efforts.



Vocational Skill

Jane Goodall began studying chimpanzees in 1960 at Gombe National Park in Tanzania. Her research demonstrated that chimpanzees are closely related to humans, not only genetically but also in behaviour. Early in her work, she observed that deforestation by local populations was destroying animal habitats. Over time, she engaged in conservation efforts, realizing that protecting their habitat was essential for chimpanzees' survival. She also noted that improving the living conditions of local communities led to cooperation in conservation efforts.

HABITAT

A **habitat** is the place where an organism lives, grows, and finds food, water, and shelter. In simple terms, it is a **dwelling place** for living beings.

Different organisms thrive in different habitats—some in mountains, forests, deserts, or water bodies. Each animal has specific characteristics and behaviours that help it survive in its habitat. These traits are known as **adaptations**.



Dwelling place : a place where an organism lives.



Different animals have different habitat.

ADAPTATION OF ANIMALS IN DIFFERENT HABITATS

Animals live in a variety of environments and rely on their physical features for survival, whether for obtaining food, staying safe, building homes, or enduring weather conditions.

Forest : A forest is a vast area covered with trees and plants, providing habitat for numerous organisms. It is home to a wide range of animals such as tigers, lions, elephants, snakes, peacocks, rats, and owls.



Desert : A desert is a dry, hot region with very little rainfall and sparse vegetation. Animals living in deserts have unique adaptations to survive in harsh conditions. Various creatures like lizards, snakes, dormice, spiders, scorpions, and insects inhabit deserts, often sheltering under rocks or in burrows.

Adaptations of Camels in the Desert:

- Camels can survive without food and water for long periods, as their bodies store both.
- They have long eyelashes, closable nostrils, and hairy ears to protect them from sandstorms.
- They conserve water by sweating less and producing minimal urine.
- Their broad, padded feet help them walk easily on sand.
- They have thick, strong mouths that allow them to eat prickly desert plants.



Polar Regions and Mountains

Polar regions have extremely cold climates. Animals like penguins, polar bears, and seals have thick fur and blubber to keep them warm. The colder regions of the world, including Canada, Siberia, and Antarctica, experience polar climates.



Prickly : covered in small, sharp thorns

Oceans

Oceans are vast bodies of salt water, covering about 71% of the Earth's surface. They provide habitat to marine life such as whales, octopuses, sharks, and various fish species.



Did You Know

Butterflies in the Amazon are known to drink tears from the eyes of turtle.

Freshwater

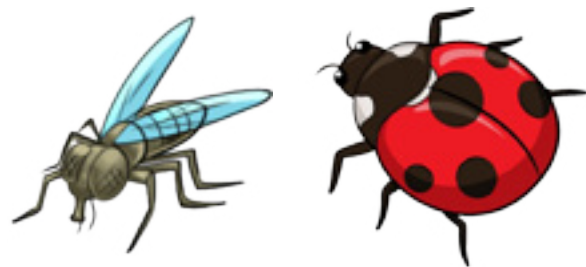
Water bodies like rivers, lakes, and ponds are known as **freshwater habitats**. Animals such as fish, frogs, and salamanders thrive in these environments. **Amphibians**, like frogs and salamanders, can live both in water and on land.



RESPIRATION IN ANIMALS

All living beings need air to breathe and survive. The process of inhaling oxygen and exhaling carbon dioxide is called respiration. Different animals have different respiratory systems:

- Humans and land animals (e.g., dogs, cows, horses, camels, lions, and tigers) breathe through lungs.
- Birds and reptiles also rely on lungs for **respiration**.
- Insects have tiny holes called spiracles on their bodies, which lead to a network of tubes called the trachea that distribute oxygen.
- Fish, crabs, and prawns breathe through gills, extracting oxygen from water.
- Amphibians (e.g., frogs and salamanders) use lungs on land and skin for respiration in water.
- Earthworms breathe through their moist skin, while amoebas use their cell membrane for respiration.



Insects have spiracles to breathe



Most of the water animals have gills to breathe



Amoeba and earthworms breathe through the surface of their bodies.



frog breathe through their lungs on land and through their skin in water.

EATING HABITS OF ANIMALS

Different animals have different feeding habits. This helps in keeping a balance in the population of living beings and in reducing the competition for the same kind of food.

Herbivores

The animals that eat only plants and plant products are called **herbivores**. They have sharp front teeth for cutting and biting leaves and grass. For example, a cow, goat,

? Did You Know
Horses can only breathe through their noses, allowing them to graze while still breathing.



Herbivores

giraffe, sheep, horse, deer, etc. Their digestive system only supports the digestion of plant-based food.

Carnivores

The animals that eat the flesh of other animals are called **carnivores**. They have sharp, pointed teeth to tear the meat. They also have strong grinding teeth to chew the meat and bones. For example, a lion, a tiger, etc. Their digestive system only supports the digestion of flesh.



Carnivores



Omnivores

Omnivores

The animals that eat both plants and flesh are called **omnivores**. For example, crows, apes, bears, rats, etc. Humans are also omnivores. Their tooth structure and digestive system support both plant-based and flesh food.

MOVEMENT IN ANIMALS

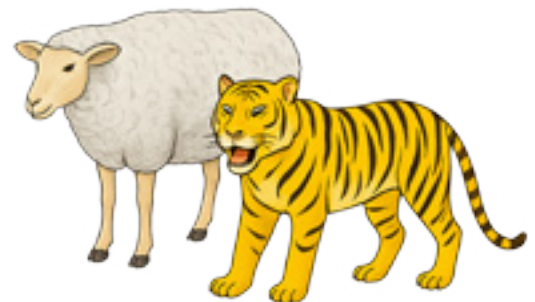
Movement in an organism is called **locomotion**. Animals move from one place to another place for many reasons. These reasons include food, water, shelter and safety. This movement can be in the same region of their habitat or even in a region hundreds to thousands of miles away from their original habitat.

All organisms like humans, animals, birds, insects, etc., move in different ways. They walk, hop, jump, crawl, slide, glide, fly, swim and climb.

Humans can stand and walk on two feet. Animals like cows, tigers, sheep, horses, deer have four limbs or feet to walk and run. The two front limbs are called fore limbs and two back limbs are called **hind limbs**. The limbs of camels are long and they have broad

? Did You Know

Butterflies in the Amazon are known to drink tears from the eyes of turtles.



Four Limbs

paddy feet which makes it easy for them to walk on sand. Horses have long and strong legs, and a single hoof which supports them to run fast with their heavy weight. The cheetah is the fastest moving animal. Its flexible body, strong muscles, gripping claws and long tail help him run so fast. Their feet have pads below them, so it does not make a noise while preying.

- Some reptiles like lizards, crocodiles, etc., have legs with which they crawl on the ground.
- Snakes are also reptiles but they do not have legs. They crawl with the help of the **scales** on their lower body.



scales: : small plate-like formation on skin



Reptiles

- Insects have six legs. Some insects also have wings. Insects like ants and cockroaches crawl.



Cockroach



Ant

- Water animals have special features to help them move under water. Fish swim by using their fins and tails. Their bodies are designed to cut through water. Animals like tortoises have paddle-like feet to move ahead in water. Frogs can move on land as well as in water.
- Most of the birds can walk on land and fly in the air. They have wings to fly and legs to walk. They have light bodies which help them fly easily. For example, an eagle, pigeon, parrot, etc. Some birds cannot fly. Emu, kiwi and ostrich are



flightless birds. Their wings are weak but their legs are very strong. They can run faster than men.



Flying birds



Flightless birds

MIGRATION

When the weather condition of a place become intolerable, many animals move to a suitable environment. The movement of large groups of animals from one place to another place is called migration. Among all animals, birds are a greatest migratory creatures. Their migration is a regular seasonal movement. They migrate thousands of miles to reach a better place for feeding and breeding. They move from colder regions to warmer regions like wildebeest and whales also migrate.



Migratory Birds

Quick Recap

- * An animal's living space is known as its habitat.
- * The unique characteristics that enable animals to survive in their habitat are called adaptations.
- * Land animals breathe using lungs.
- * Insects take in air through tiny openings called spiracles.
- * Fish use gills for breathing.
- * Herbivores consume only plants, while carnivores feed solely on meat.
- * Omnivores eat both plant-based and animal-based food.
- * Many animals, including humans, use their legs for movement.
- * Reptiles move by crawling.
- * Birds use their wings to fly.
- * Among all creatures, birds are the most remarkable migrators.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. The environment where an animal lives is called its:
a) Home ☐ b) Habitat ☐
c) Shelter ☐
2. Frogs breathe on land using their:
a) Gills ☐ b) Lungs ☐
c) Skin ☐
3. Which animal moves with the help of its scales?
a) Sparrow ☐ b) Housefly ☐
c) Snake ☐
4. Insects breathe through body openings known as:
a) Stomata ☐ b) Lungs ☐
c) Spiracles ☐

B. Answer in One Word

1. The two hind limbs of an animal _____
2. The “ship of the desert” _____
3. A bird that cannot fly _____
4. Openings on an insect’s body that allow breathing are called _____
5. An animal that breathes through both lungs and skin _____

C. Write T for true and F for false statements.

1. Camels have paddy feet. ☐
2. Lizards and dormice are found in hilly regions. ☐
3. Crabs breathe through their skin. ☐
4. Flesh-eating animals are called herbivores. ☐
5. Kiwis can fly faster than pigeons. ☐

D. Answer the Following Questions

1. What is a habitat? List different types of habitats.
2. What is adaptation? Give examples of animals adapted to desert life.
3. Differentiate between herbivores and carnivores.
4. How do insects breathe?
5. What is migration? Why do animals migrate?

CAPABILITY ANALYSIS

A. Scrapbook Activity

Art Integration

Gather images of different animals' feet, paste them in a scrapbook, and explain how their feet help them survive in their habitat.

B. Camel Adaptation Study

Critical Thinking

Observe a picture of a camel and describe how its body parts help it adapt:



1. Eyelashes

2. Nostrils

3. Hump

4. Legs & Feet

C. Poster Making Design a poster on “Save Animals” to promote wildlife conservation.

Art Integration

D. Wildlife Sanctuary Visit If a wildlife sanctuary or national park is near your town, visit it to observe animals in their natural habitat.

Experiential Learning

E. Research on Valmik Thapar Valmik Thapar is a renowned Indian natural historian known for his work in tiger conservation. He has authored books like *The Tiger: A Destiny and Encounter* and contributed to wildlife documentaries for BBC, Animal Planet, and National Geographic. Research his work online.



Digital Resources

F. Feather Art Collect large feathers, paint them in vibrant colours, and display them in a vase at home.

Vocational Skill



G. Internet Research on Animals Find information about different animals, including:

Digital Resources

1.



Animal Name: _____

Natural Habitat: _____

Movement: _____

Eating Habit: _____

Respiration: _____

2. Animal Name: _____

Natural Habitat: _____

Movement: _____

Eating Habit: _____

Respiration: _____



3.



Animal Name: _____

Natural Habitat: _____

Movement: _____

Eating Habit: _____

Respiration: _____

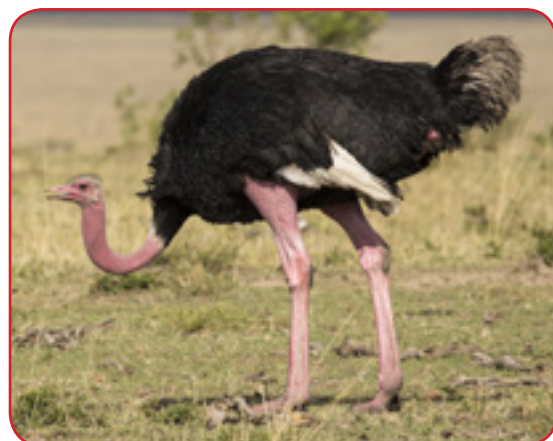
4. Animal Name: _____

Natural Habitat: _____

Movement: _____

Eating Habit: _____

Respiration: _____



H. Sustainable Development Goals Goal 4: Quality Education – Learn about the importance of protecting animals and their habitats.

Creativity

HOTS

Which animals can swim and fly? How do their bodies support movement in both air and water?

Scientific Temper

Examples include ducks, swans, and penguins.

They have webbed feet for swimming and strong wings for flying. Their lightweight bones help them stay buoyant in water and agile in the air.

Skeletal System and Nervous System

Objectives



- *Skeletal System*
- *Nervous System*

Our body is made up of various organs such as the heart, brain, mouth, and stomach. However, a single organ cannot function on its own. A collection of organs that collaborate to carry out a specific function is known as an **organ system**. The human body has multiple organ systems.

How are you feeling right now? Do you know how to process your emotions? It's normal to experience different feelings—what matters is how you express them.

Critical Thinking

I Feel....

**Angry, Annoyed,
Terrified, Stressed**



**Confused, Worried,
Nervous, Frightened**



Sick, Tired, Hurt, Sad



**Calm, Excited, Focused,
Happy**



I need to....

Breathe deeply, count to ten, and ask for help.

Take a break, get fresh air, and talk to someone.

Rest, engage in an enjoyable activity, and speak with an adult.

Embrace and share my positive emotions.

SKELETAL SYSTEM

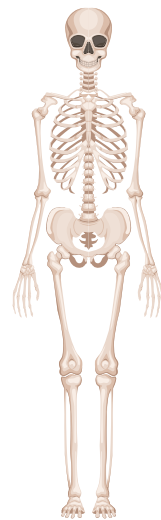
Let's explore the skeletal and muscular systems in detail.

The Skeleton

Can you imagine life without a skeletal system? Without it, we wouldn't be able to sit, stand, or move.

The skeletal system forms the body's framework, giving it structure and support. It also safeguards vital organs and provides strength. The human skeletal system consists of 206 bones, varying in size from large to tiny. These bones are made of living cells.

Although bones are hard, they contain small holes that allow blood vessels and nerves to pass through. Within bones, **bone marrow**, a fatty substance found in long bones, produces blood cells.



Skeletal System



Did You Know

The smallest bone in the human body is the stirrup bone, located in the ear.

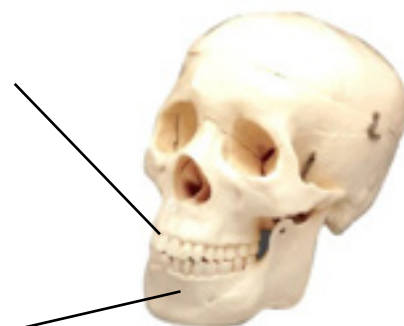
The Skull

The skull protects the brain and consists of 22 bones.

The upper jaw is immovable, while the lower jaw can move, enabling us to chew and speak.

Upper jaw

Lower jaw



The Backbone (Spine)

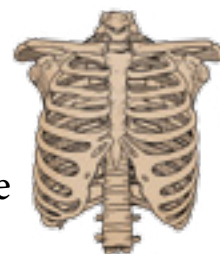
The backbone shields the spinal cord, nerves, and blood vessels.

It is the most crucial bone in the body.

It consists of 33 small bones called **vertebrae**, forming the body's central support structure.

The Rib Cage

The ribs are curved bones connected to the backbone at the rear and the sternum at the front.



Floating ribs

Together, they form a protective **rib cage**, safeguarding the heart and lungs. There are 12 pairs of ribs. The last two pairs, called **floating ribs**, are only attached to the backbone.

? Did You Know

If a bone fractures, it heals by producing new cells and blood vessels.

The Limbs

Humans have two pairs of limbs:

Upper limbs (arms) The upper arm has one bone, while the lower arm has two bones. The wrist, palm, and fingers contain multiple bones.

Lower limbs (legs) That high bone is the longest bone in the human body. The lower leg has two bones connected at the knee.

Joints

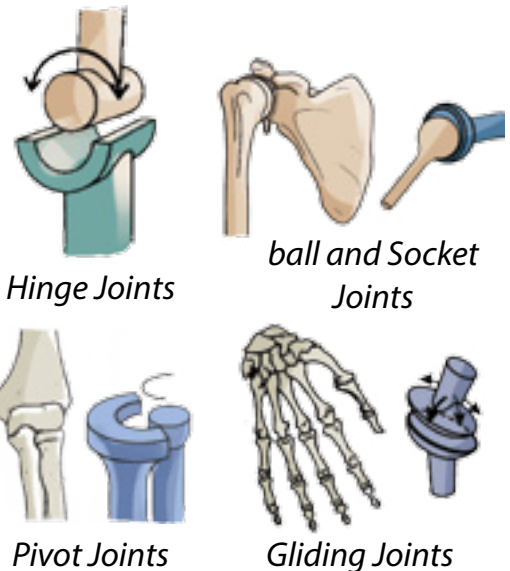
A **joint** is a point where two bones meet. **Ligaments**, strong connective tissues, hold bones together. There are various types of joints:

Hinge Joints Allow movement in one direction (e.g., elbow, knee).

Ball-and-Socket Joints → Allow movement in many directions (e.g., hip, shoulder).

Pivot Joints Enable side-to-side and up-and-down movement (e.g., between the skull and the spine).

Gliding Joints Allow bones to slide past one another (e.g., wrist, ankle).



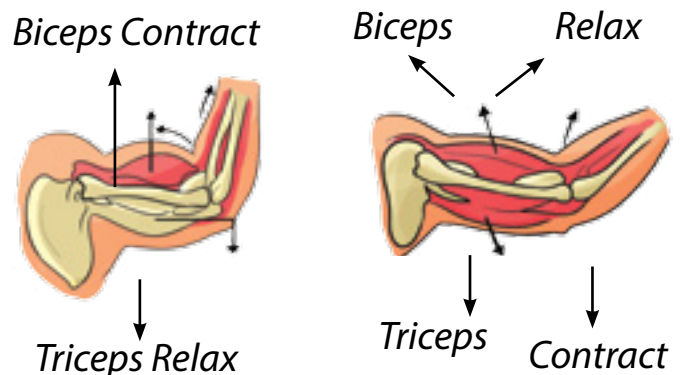
Muscles

Muscles are soft tissues that cover bones and facilitate movement. The body has approximately 650 muscles responsible for breathing, digestion, and movement.

Muscles work in pairs When one contracts, the other relaxes.

Tendons Connect muscles to bones.

Muscles help with Breathing, eating, digestion, and various movements.



Pulling up our Arm Lowering down our Arm

Types of Muscles

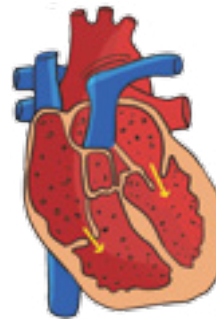
Voluntary Muscles Controlled by our will (e.g., biceps, triceps).

Involuntary Muscles Function without conscious control (e.g., muscles aiding digestion and breathing).

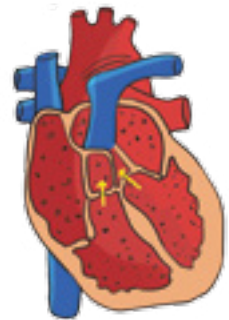
Cardiac Muscle Found in the heart, working continuously until death.

Did You Know

A smile uses 14 muscles. Keep smiling!



Expansion



Contraction

The heart contracts and expand to make the blood flow.

How Do Muscles Work?

Muscles pull bones to produce movement.

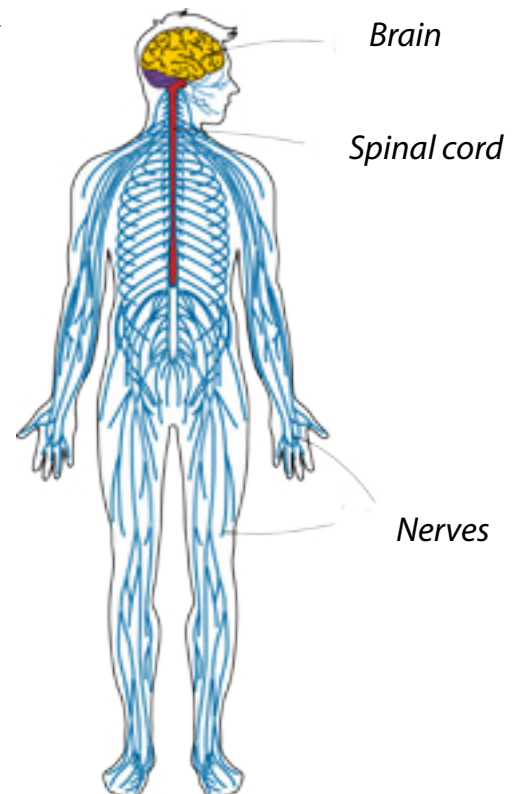
When the knee bends, one muscle contracts while the other relaxes, and vice versa when the leg stretches.

Did You Know

Cardiac and voluntary muscles have striations (light and dark bands).

NERVOUS SYSTEM

The nervous system is crucial for controlling body functions and actions, such as walking, talking, and thinking. It consists of three main parts: the brain, the spinal cord, and the nerves. These parts work together to receive information, process it, and send messages to different body parts. The brain acts like the control centre, while the spinal cord and nerves carry signals quickly. This system helps the body respond to changes inside and outside the environment.



The Nerves System

Did You Know

Sense organs collect information and send it to the spinal cord, which then transfers it to the brain.

The Brain

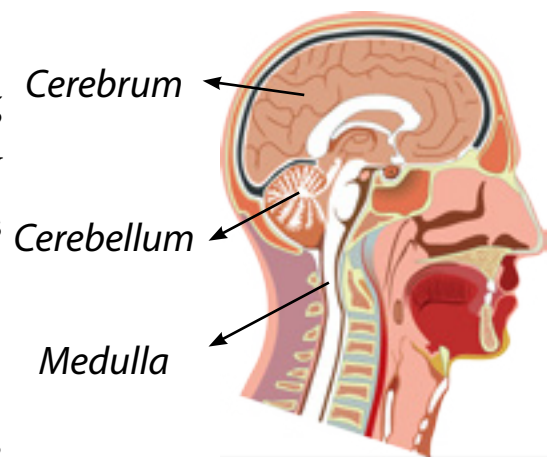
The brain is the body's control centre, receiving signals from all parts via nerves. It is protected by the skull and cushioned by a clear fluid that absorbs shocks.

Three major parts of the brain:

Cerebrum The largest part, controlling senses, memory, speech, and thoughts.

Cerebellum The small brain, regulating muscle movement.

Medulla Manages breathing, heart rate, and involuntary actions (e.g., sneezing).



The Brain

? Did You Know

A human brain weighs approximately 1.4 kg.

The Spinal Cord and Nerves

The spinal cord is a thick bundle of nerve tissue that extends from the brain and connects with all body parts.

It is safeguarded by the **vertebral column** and controls **reflex actions**.

Types of Nerves

Sensory Nerves : Carry messages from sense organs to the brain or spinal cord.

Motor Nerves : Transmit commands from the brain or spinal cord to muscles.

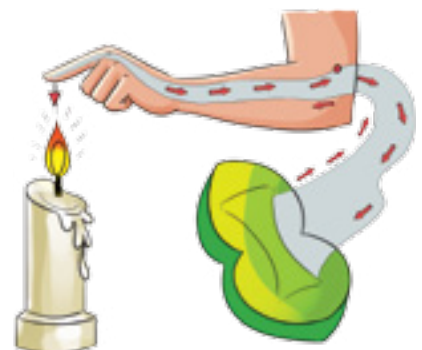
Mixed Nerves : Perform both sensory and motor functions.



Spinal Cord

REFLEX ACTIONS

A reflex action is an instantaneous reaction to an external stimulus. Example: If you touch something hot, your hand moves away immediately, controlled by the spinal cord, not the brain.



Reflex Actions

Quick Recap

- * The skeletal system provides structure and support.
- * The skull, rib cage, backbone, and limbs are key components of the skeletal system.
- * The nervous system controls body functions and actions.
- * The brain, spinal cord, and nerves make up the nervous system.
- * Nerves transmit messages between the brain and body.
- * The spinal cord governs reflex actions.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. _____ is an example of a hinge joint.

a) Elbow

☐

b) Wrist

☐

c) Ankle

☐

2. The human backbone consists of _____ bones.

a) 34

☐

b) 33

☐

c) 35

☐

3. What does our nervous system regulates

a) Body function

☐

b) Body action

☐

c) Both a) and b)

☐

4. All movement of our body are controlled by

a) Brain

☐

b) Spinal cord

☐

c) Muscles

☐

5. Which of the following is the largest part of the brain.

a) Cerebellum

☐

b) Cerebrum

☐

c) Medulla

☐

B. Fill in the blanks with the appropriate words from the box.

frame work cerebellum reflex action medulla involuntary

1. The skeletal system serves as the body's _____
2. Muscles that function independently are known as _____.
3. _____, a bulb-shaped structure, links the brain and spinal cord.
4. The _____ is often referred to as the “small brain.”
5. A _____ is the body's automatic response to a stimulus.

C. Write T for true and F for false statements.

1. The hip joint is a hinge joint.
2. The skull consists of eight flat bones.
3. Tendons connect bones together.
4. The human body contains over 650 bones.
5. The heart is a voluntary muscle.

☐
☐
☐
☐
☐

D. Match the Columns.

Column A

1. Cerebrum
2. Vertebrae
3. Cerebellum
4. Triceps
5. Cardiac Muscles

Column B

- (a) Backbone
- (b) Voluntary muscle
- (c) Largest part of the brain
- (d) Involuntary muscle
- (e) Small brain

E. Answer the Following Questions.

1. Name the different types of Joints in the Human Body?
2. What is the distinction Between Voluntary and Involuntary Muscles?
3. State the different parts of the Brain and define them?
4. Explain the Spinal Cord?
5. State the definition and types of Nerves?
6. What is Reflex Action and how does it work? Define with an example?

CAPABILITY ANALYSIS

A. Discussion on “The Brain as the Centre of Intelligence”

Communication Skill

B. Researching and Drawing the Braille Alphabet:

Multidisciplinary Learning

Louis Braille was himself blind, and he invented the Braille script to help blind people read and write. He lost his eyesight at the age of three due to an accident. With great determination, he created a system of raised dots that gave hope and light to millions of visually impaired people across the world.



C. Recognizing Reflex Action Pathway:

Multidisciplinary Learning

D. Observing Exercise Techniques in a Gym:

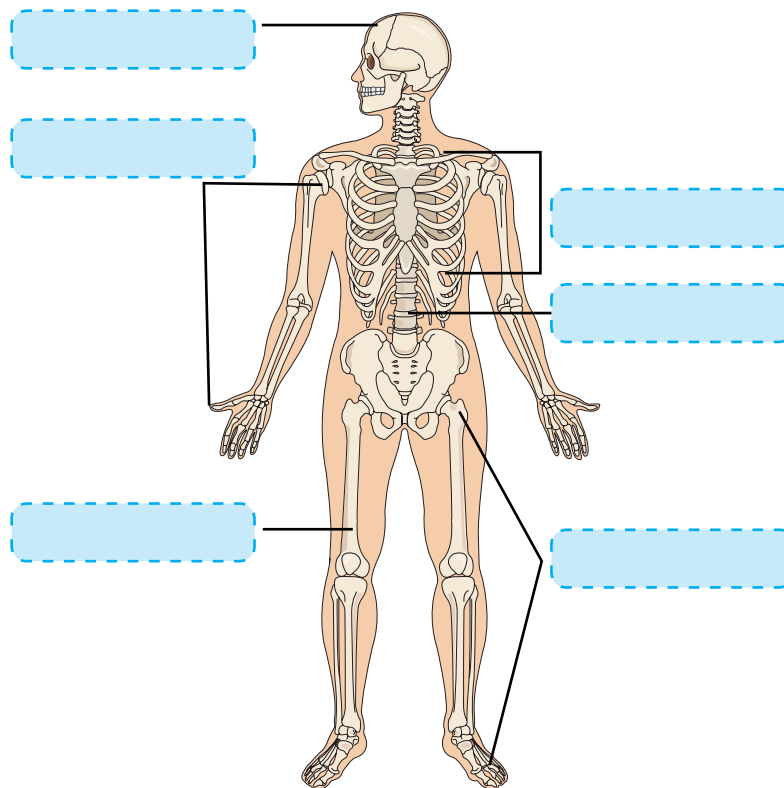
Experiential Learning

E. Creating an Informative Chart on Sense Organs:

Art Integration

F. Label the Skeletal System with Given Terms:

Critical Thinking



HOTS

What if the backbone were a single bone instead of multiple vertebrae?

Critical Thinking

Force, Work, and Energy

Objectives



- Force

- Types of Energy

- Types of Forces

- Work

- Energy

Energy Conservation: Small Steps, Big Impact Have you ever considered how small changes in our homes can contribute to a healthier environment? By adopting simple energy-saving habits, we can not only protect natural resources but also lower our electricity bills.

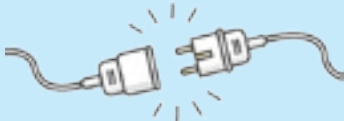
Here are some effective ways to reduce energy consumption at home

Environment and Health

- 

Switch off lights when leaving a room.
- 

Use energy-efficient appliances.
- 

Replace regular bulbs with LED lights.
- 

Unplug devices when they are not in use.
- 

Set the thermostat at a lower temperature.
- 

Reduce water usage.
- 

Use smart automation for better efficiency.
- 

Install double-glazed windows.
- 

Cook food with the lid on to conserve heat.
- 

A smart meter tracks energy use.
- 

Wash clothes at a cold temperature.
- 

Use for solar-powered devices.

FORCE

A **force** is a push or pull applied to an object. It plays a crucial role in motion, whether moving an object, stopping it, or changing its speed or direction. You use force in daily activities such as walking, lifting, or throwing objects.

Effects of Force:

- It can stop a moving object.
- It can set a **stationary** object in motion.
- It can increase the speed of a moving object.
- It can slow down a moving object.
- It can change the direction of an object in motion.
- It can alter the shape or size of an object.



Stationary : Not in motion



Pushing a door open



Writing on a piece of paper



Playing hockey



Kicking a football



Running as an exercise



Pulling a suitcase

TYPES OF FORCES

Mechanical Force: This type of force occurs when objects make direct contact. For example, using scissors to cut fabric or a knife to chop vegetables requires mechanical force. It involves pushing or pulling an object by touching it. Everyday tasks like writing with a pen or opening a door use mechanical force.



Mechanical Force



Muscular Force

Muscular Force: This force is exerted by muscles when we push, pull, or lift objects. Walking, running, and lifting weights are all examples of muscular force.



Exerted : To apply force or strength

Gravitational Force: The force that pulls objects toward the Earth. For instance, when a ball is thrown upwards, it eventually falls back due to gravity.



Gravitational Force



Frictional Force

Frictional Force: This is the resistance that occurs when two surfaces move against each other. For example, a rolling ball gradually slows down due to friction.

Magnetic Force: A magnet attracts iron objects due to magnetic force. This force arises from the movement of electric charges.



Magnetic Force



Electrostatic Force

Electrostatic Force: A charged object can exert force on another charged or uncharged object. For example, rubbing a comb through dry hair makes it attract small paper pieces.

Elastic Force: When a stretched rubber band or spring is released,

it returns to its original shape due to elastic force.



Elastic Force

? Did You Know

Friction generates heat, which can be both useful and harmful.

Buoyant Force: When an object is placed in water, it experiences an upward force that helps it float. This force is called buoyant force.



Buoyant Force

ENERGY

Energy is the ability to perform work. Every activity, from playing and studying to eating and moving, requires energy.



Thrust : A strong push

TYPES OF ENERGY

Solar Energy: The heat and light from the sun provide solar energy. It is used in solar cookers, solar water heaters, and solar-powered devices like watches and calculators.



Solar Energy

Heat Energy: This energy is produced when substances like wood and coal are burned. The sun is the primary source of heat energy.

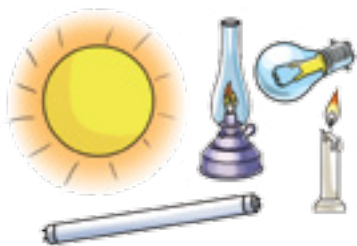


Heat Energy



Did You Know

The food you eat provides chemical energy to your body.



Light Energy

Light Energy: The energy emitted by sources such as the sun, bulbs, candles, and lamps. Light energy is composed of photons, which are generated when atoms heat up.

Electrical Energy: Appliances such as refrigerators, TVs, and washing machines operate using electrical energy, which is transmitted through power lines.



Electrical Energy



Did You Know

Electric vehicles (EVs) are an eco-friendly alternative to fuel-powered cars. Unlike conventional vehicles, they run on electricity, reducing pollution and costs.



Kinetic Energy: The energy possessed by a moving object. Examples include a moving bicycle, car, or running person. This energy is called kinetic energy.



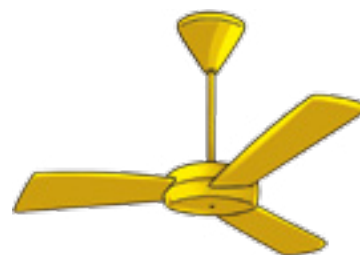
Kinetic Energy



Potential Energy

Potential Energy: Stored energy based on an object's position. For instance, a stretched rubber band or a bicycle on top of a hill contains potential energy. It is released when the object changes its position or shape.

Mechanical Energy: A combination of kinetic and potential energy. For example, a moving ceiling fan has **mechanical energy**.



Mechanical Energy



Wind Energy

Wind Energy: Wind turbines convert the kinetic energy of wind into mechanical energy, which can then be used to generate electricity. Energy is never created; it only changes forms. For example, burning coal releases heat and light energy, which can be converted into electrical energy.

Did You Know

You get energy when you eat food, this energy is called chemical energy.

WORK

Work is done when a force is applied to an object, causing it to move in the same direction as the force. In simple terms, work involves moving an object from one position to another using force.

For example, if you push a heavy object and it does not move, no work is done. However, if you pull a chair and it changes position, work has been accomplished.



Constant : Continuous or unchanging

Quick Recap

- * Force is a push or pull applied to an object.
- * Energy is the ability to perform work.
- * Different types of forces and energy exist.
- * Energy is not created but transformed from one form to another.
- * Work occurs when force causes displacement in the same direction.
- * Power and energy are essential for performing any task.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

- The force exerted by a charged body on another object is called ____ force.
a) Magnetic ☐ b) Electrostatic ☐
c) Muscular ☐
- A horse pulling a cart is an example of ____ force.
a) Muscular ☐ b) Gravitational ☐
c) Elastic ☐
- The primary source of energy on Earth is the _____.
a) Sun ☐ b) Water ☐
c) Wind ☐
- A mug floating on water experiences _____ force.
a) Elastic ☐ b) Frictional ☐
c) Buoyant ☐
- Gravitational force acts in a _____ direction.
a) Pulling up ☐ b) Pulling down ☐
c) Pulling sideways ☐

B. Fill in the blanks with the appropriate words from the box.

Solar cookers

Magnetic

Work

Change

Force

1. A push or pull is called _____
2. Force can _____ the shape of an object.
3. _____ force attracts iron nails.
4. Solar energy is used to cook food in _____
5. _____ occurs when force causes displacement.

C. Provide two examples of each.

- | | | |
|-----------------------|-------|-------|
| 1. Magnetic Force: | _____ | _____ |
| 2. Frictional Force: | _____ | _____ |
| 3. Elastic Force: | _____ | _____ |
| 4. Light Energy: | _____ | _____ |
| 5. Electrical Energy: | _____ | _____ |

D. Write T for true and F for false statements.

1. Force can move a stationary object.
2. Magnetic force is caused by moving charges.
3. Frictional force opposes motion.
4. Gravitational force attracts only iron objects.
5. Buoyant force helps objects float in water.

☐
☐
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☐

E. Answer the following questions.

1. What is force?
2. What are four effects that force can have on objects?
3. What is energy? List three types of energy.
4. How is light energy different from heat energy?
5. What is work? Provide an example.

CAPABILITY ANALYSIS

A. Observe the picture and respond to the following questions

Scientific Temper

1. What is the name of the structure shown in the image?

2. Is it a renewable or non-renewable energy source? Justify your answer.

3. What form of energy does it utilize?



B. Identify the type of force depicted in the following images:

Critical Thinking

1.



2.



3.



C. Group Activity: Understanding Different Energy Sources

Collaboration

Divide the class into three groups, each focusing on one of the following energy Sources: Solar Energy, Hydroelectric Energy, and Electrical Energy. Each group will research and gather images related to their assigned energy source to create a chart, then present their findings and answer questions from peers.

D. Hands-on Exploration: Force and Motion

Multidisciplinary Learning

Participate in engaging activities to explore force and motion through interactive experiments. Integrate these principles with other subjects to develop a deeper understanding and observe different types of forces and their impact.

PUSH & PULL PAINTING

Explore the force of push and pull as you try to create a unique abstract work of art.



Materials required: White Card stock paper, tape, tray, small size cardboard rectangles and paint.

Method: : Participate in activities to explore force and motion through experiments. Integrate these principles with other subjects to understand different forces and their impact.

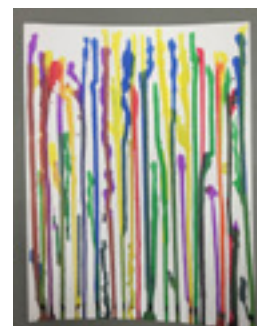


GRAVITY PAINTING

Gravity is what keeps people and objects on the ground and causes objects to fall towards the Earth. Use the Earth's gravitational pull to direct the paint onto your canvas.

Materials required: Cardboard, paint, eyedropper and drop cloth.

Method: Prep your workspace with a drop cloth. Prop your board against a table or easel. Use an eyedropper to drip the paint at the top of your board. Experiment with manipulating the drips by turning the board as the drips run down.

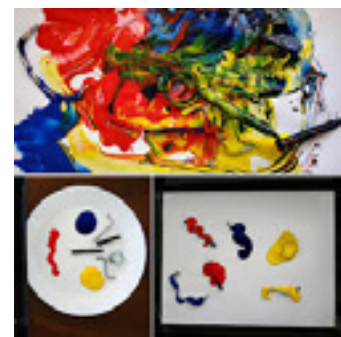


MAGNETIC PAINTING

Explore magnetic force with this fun hands-on painting activity.

Materials required: Small metal objects, magnet, card stock paper, tape, washable paint and tray.

Method: Tape a piece of white paper to the sheet. Place small metal objects on it and a magnet underneath. Move the objects around the paper with the magnet, then add paint and guide them through it. Try moving them fast, slow, or in circles until satisfied with the design.



HOTS

Four blocks are placed at the edge of a table as shown. Which block will fall over? State the reason.



Scientific Temper

SIMPLE MACHINES

Objectives



- Understanding Simple Machines
- Types of Simple Machines
- Exploring Complex Machines

A machine makes tasks easier, especially when doing something difficult by hand.

Some simple household machines are shown. Can you identify them? Match them with their uses.

Critical Thinking



SIMPLE MACHINES

In everyday life, you perform various tasks that require specific tools. For example, moving a heavy stone by hand is tough, but using a long rod makes it easier. Tools that assist in making work simpler and faster are called **simple machines**. These machines have little to no moving parts but help us perform tasks efficiently.



Screwdriver



Tongs



Lever



Wedge



Screw

TYPES OF SIMPLE MACHINES

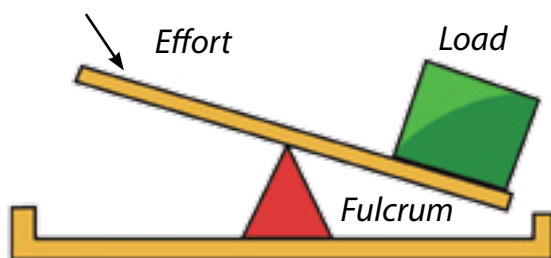
There are six main types of simple machines:

Lever

A **lever** is a simple machine consisting of a rigid bar that pivots on a fixed point called a **fulcrum**. It helps **amplifies** force, making it easier to lift or move objects. A lever consists of three main parts: Load (L): The object being moved. Effort (E): The force applied to move the **load**. Fulcrum (F): The point of support on which the lever pivots.

There are three classes of levers:

1. **First-Class Lever:** The fulcrum is positioned between the load and **effort**. Examples include scissors, claw hammers, and pliers.

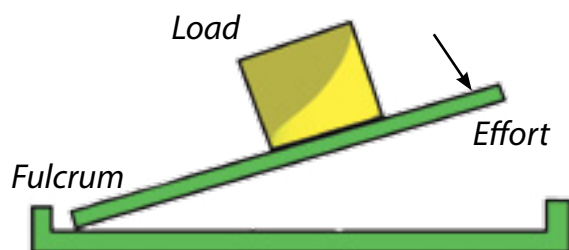


scissors



pliers

2. **Second-Class Lever:** The load is positioned between the fulcrum and the effort. Examples include a wheelbarrow, bottle opener, and nutcracker.



Wheelbarrow



Bottle opener

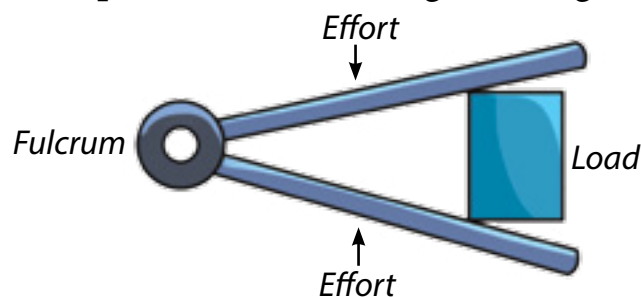


Nutcracker

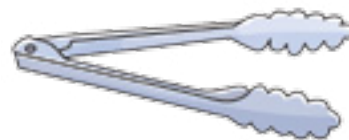


level	: a rigid rod
fulcrum	: pivot of a lever
amplifies	: multiplies

3. **Third-Class Lever:** The effort is applied between the fulcrum and the load. Examples include ice tongs, fishing rods, and forceps.



Stapler



Tongs

Inclined Plane

An **inclined plane** is a sloping surface that makes it easier to move heavy objects up or down. Instead of lifting an object straight up, you can push it along the slope, requiring less force. Examples: Sloping wooden planks, ramps, and slides. In hospitals, ramps are preferred over stairs for moving patients conveniently.



Inclined : A surface that slopes from a higher point to a lower point.



Screw

A **screw** is an inclined plane wrapped around a cylindrical shape. The pointed end is called the **tip**, while the opposite end is the head. Screws are commonly used to hold materials together. Examples: Screws in woodwork, electric bulbs, clamps, jacks, key rings, and spiral staircases.

Inclined plane



Wheel and Axle

A **wheel and axle** is a simple machine where a wheel is connected to a rod (axle). When one moves, the other moves as well. This system helps lift or move objects with ease. Examples: Steering wheels, cars, sewing machines, and egg beaters.



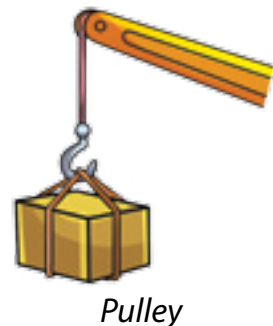
Wheel and Axle



Screw

Pulley

A **pulley** is a wheel with a rope or cable wrapped around it. One end of the rope is attached to a load. When you pull on one side of the rope, the wheel turns, and the load moves in the opposite direction. Pulleys help lift objects by changing the direction of force, making heavy lifting easier. Example: Cranes use pulleys to lift heavy materials.



Pulley

Wedge

A **wedge** is a double inclined plane that tapers to a sharp edge on one side. The sharp edge helps cut or split objects apart like: knives, axes, and shovels.



Axle : A rod that passes through a wheel.



Axe



Shovel



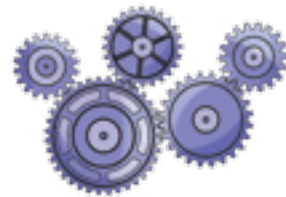
Needle



Knife

Gear

A gear is a toothed wheel that interlocks with another gear. When one gear turns, it causes the other to turn, allowing control of speed and direction. Example: Gears in mechanical clocks.



Gear

COMPLEX MACHINES

A **complex machine** consists of two or more simple machines working together. These machines are also known as **compound machines**. Most of the devices we use daily are complex machines. Examples: Cars, washing machines, staplers, hand drills, and cranes. Some machines are even larger and more advanced, such as bicycles, cars, washing machines, and cranes.



Typewriter



Crane



Lawn mower



Drilling machine



Quick Reflection

1. First-Class Lever: _____
2. Wedge: _____
3. Inclined Plane: _____
4. Complex Machines: _____
5. Simple machine: _____

Quick Recap

- * Simple machines are tools with minimal moving parts that make work easier.
- * A lever is a rigid bar that pivots on a fulcrum.
- * An inclined plane is a sloping surface.
- * A screw is a cylinder with an inclined plane wrapped around it.
- * A pulley consists of a wheel and a rope or chain that moves over it.
- * A wedge is a tool with two inclined planes meeting at a sharp edge.
- * A complex machine is a combination of multiple simple machines.
- * Examples of complex machines: Bikes, cars, washing machines, cranes, etc.

ABSTRACT ANALYSIS

A. Choose the correct answer:

1. What helps make our work easier and faster?
a) Hand machines ☐ b) Small machines ☐
c) Simple machines ☐
2. The point where a lever pivots is called the:
a) Fulcrum ☐ b) Turning point ☐
c) Resting point ☐
3. A bottle opener is a:
a) First-class lever ☐ b) Pliers ☐
c) Second-class lever ☐
4. A screw is a cylindrical machine with an inclined plane wrapped around it.
a) Lever ☐ b) Screw ☐
c) Gear ☐

5. The rod that passes through a wheel is called:

a) Axle

☐

b) Lever

☐

c) Wedge

☐

6. A hospital ramp is an example of:

a) Pulley

☐

b) Lever

☐

c) Inclined plane

☐

B. Fill in the blanks with the appropriate words from the box.

inclined plane second class lever mechanical third class lever first class lever

1. Simple machines use _____ force.

2. A sea-saw in a park is an example of a _____.

3. Sloping wooden planks and slides are examples of an _____.

4. A _____ has the load between the fulcrum and the effort.

5. A _____ has the effort between the load and fulcrum.

C. Write T for true and F for false statements.

1. A screw is a simple machine.

☐

2. A wheel is a simple machine.

☐

3. A ramp is a smooth surface.

☐

4. A complex machine is made by combining many simple machines.

☐

5. A washing machine is a complex machine.

☐

6. A stapler is an example of a second-class lever.

☐

D. Answer the following questions.

1. What is a simple machine? How does it make work easier?

2. What is a lever? Describe its types with examples.

3. How does a pulley function? What are its uses?

4. Mention two uses of an inclined plane.

5. How is a wedge formed? How can it become a complex machine?

CAPABILITY ANALYSIS

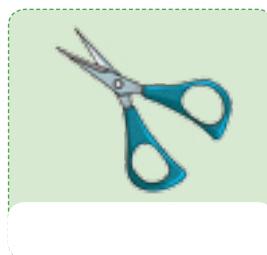
A. Classify the following as first, second, or third-class levers:

Critical Thinking

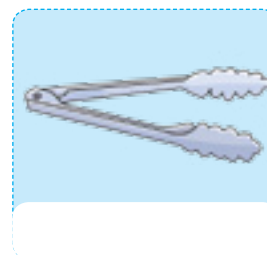
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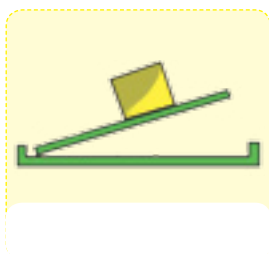
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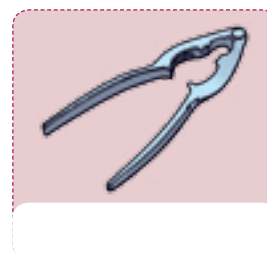
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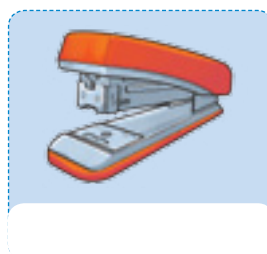
6.



7.



8.



9.



B. Identify manual and mechanical activities:

Critical Thinking

Manual Jobs (Using Hands)

1. Dusting the house

2. _____

3. _____

4. _____

Mechanical Activities (Using Machines)

1. Cutting paper with scissors

2. _____

3. _____

4. _____

HOTS

How do simple machines differ from complex machines? Is a computer a complex machine?

Conceptual Learning

Air and Water

Objectives



- Air
- Water
- Purification of Water

We need air to breathe and water to drink. Water and air are essential for maintaining life on Earth.

Make your own 'Water Filter'.

Hand-on Learning

You will need a clean plastic bottle, a glass, pebbles, gravel, and sand.

Method: Cut off the bottom of a clear plastic bottle. Stuff the neck of the bottle with clean cotton wool. Turn the bottle upside down and place it on a clean glass. Place cotton inside the bottle. Over the cotton wool, spread a layer of small pebbles, then gravel, and above that, a layer of sand. All pebbles and gravel must be washed before putting them in the bottle.

Pour dirty water over these layers of filters. Watch the water drops trickling down through the cotton wool. It is clean water. This way, dirty water is being filtered and all the insoluble impurities are removed from the water. Remember, this water is still not fit for drinking. Find the reason!



AIR

Air is present all around you. You cannot see or smell air. However, you can feel it when it moves. Air is essential for life. You use it to get oxygen which is needed for breathing. Plants take carbon dioxide from it to make food.

Earth is the only planet where life exists. Life is not possible on the moon as there is no air at all. It is proved that there is no air on the moon as the footprints of all the

astronauts who visited there are still intact.



Atmosphere

Our Earth is surrounded by a blanket of air called **atmosphere**. It extends up to several hundreds of kilometers above the Earth. The region outside the atmosphere where there is no air is called **space**.

The atmosphere gives you oxygen to survive. It gives CO₂ for plants' survival. It maintains the temperature on Earth. It burns meteorites before they reach the Earth's surface. The atmosphere makes a safety layer (**ozone layer**) to protect the Earth from many harmful rays like **ultraviolet rays**.



Composition of Air

Air is a mixture of many gases. It contains about 78% Nitrogen, 21% Oxygen, and 1% other gases.

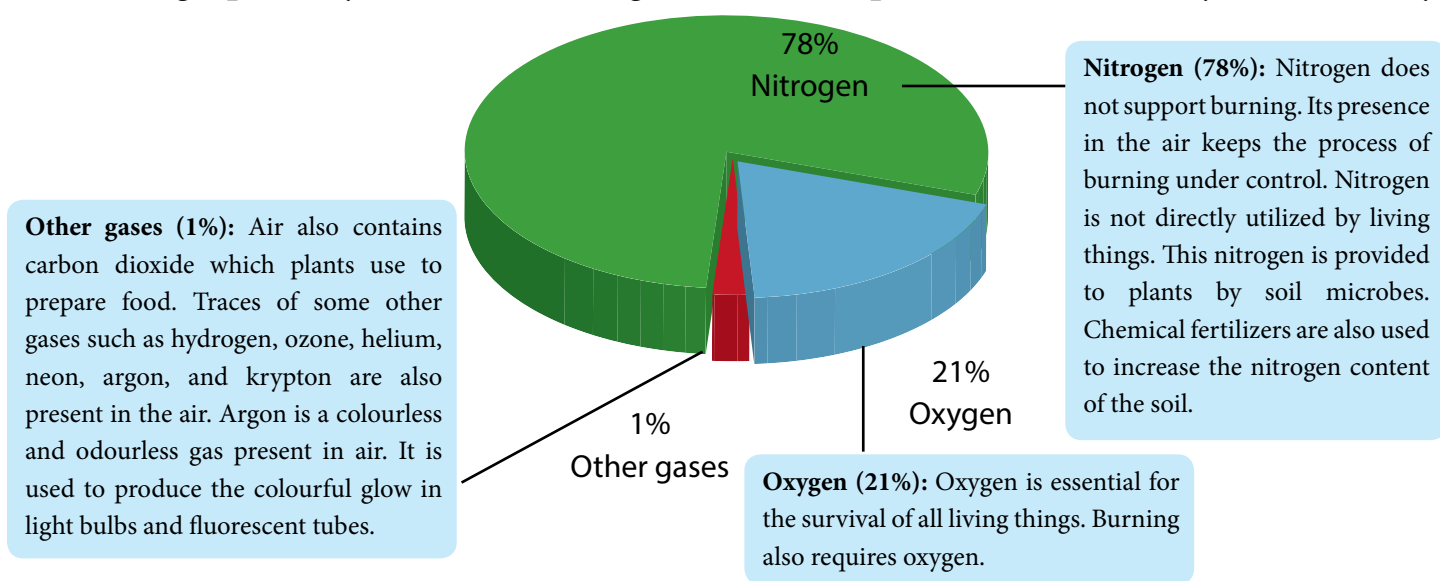
The other gases that constitute 1% are carbon dioxide, argon, hydrogen, helium, and neon. There is a variation in the percentage of dust, smoke, water vapor, and air pollutants from place to place.

- Oxygen is the most important gas in air. All living beings require oxygen to

breathe. It burns the food you eat to give you energy. Oxygen is also essential for combustion, so it is needed for the burning of fuel.

Nitrogen is an important constituent of the protein. So, it is needed by the plants to make proteins for you.

Carbon dioxide is essential for photosynthesis in plants. Plants make their food through photosynthesis. You all get food from plants either directly or indirectly.

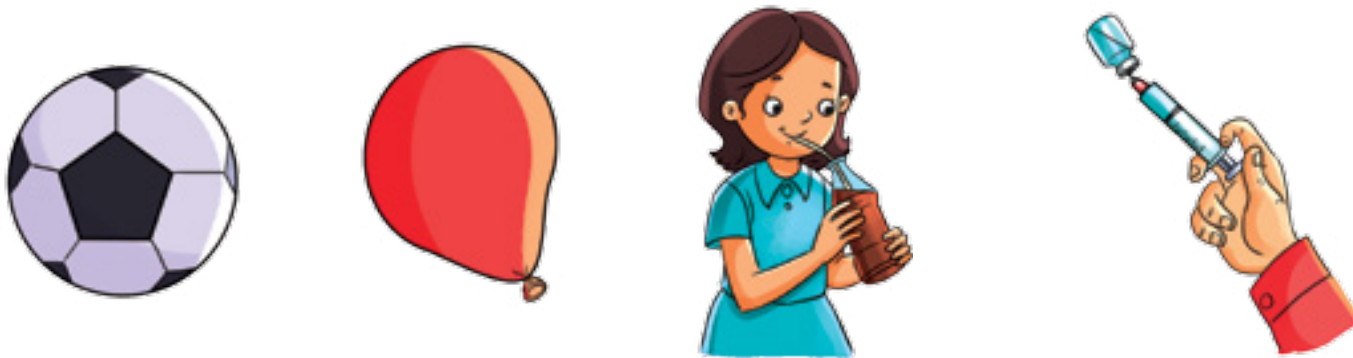


- **Water vapour** determines the condition of the weather in an area. The excess water vapour in the air is called **humidity**. If air contains more water vapour than its capacity, it is called **humid air**. During humid days, your body feels sticky and clothes take more time to dry.
- All other gases have some use, e.g., neon and argon are used in bulbs. The absence of air or any other matter at a particular place is known as a **vacuum**.

Properties of Air

- **Air occupies space:** Air takes up space because when you put air in the bicycle's tyre or balloon, it becomes bigger. This process of filling air into any expandable substance is called **inflation**.
- **Air has weight:** You compare two footballs. One of them is inflated but the other is deflated. When you hold them in your hands, you can easily feel the difference between their weights, i.e., the inflated football is heavier than the deflated one.
- **Air exerts pressure:** You have learnt that air occupies space and has weight.

Anything with weight exerts pressure, so air also exerts pressure. The pressure exerted by air on the Earth is called atmospheric pressure. It acts in all directions, even on your body, but you don't feel it as it is equal from all sides. Air pressure is useful—doctors use it in syringes, you drink with a straw, tyres are filled with air for smooth driving, and it helps in drawing water with pumps.



? Did You Know

A siphon is a bent tube through which you draw liquids from containers.
A barometer is a scientific instrument for measuring air pressure.

Activity

Fill a glass with water and place it on a table. Insert one end of a transparent tube into the glass and suck air from the other end. Keep this end in a glass at a lower level, and water will rush into it until the level in both glasses is the same.

In this experiment, a plastic tube acts as a siphon.



Air Pollution

The addition of unwanted harmful substances to the air is called air pollution. These substances are called pollutants. Main pollutants are unburnt carbon particles, carbon monoxide, sulphur dioxide, nitrogen dioxide and lead.



Air Pollution

Sulphur dioxide and nitrogen dioxide are very harmful as they cause acid rain, which damages plants and buildings.

AQI

Good
0-50
Moderate
51-100
Unhealthy for
sensitive groups
101-150
Unhealthy
151-200
Very unhealthy
201-300
Hazardous
301-350



AQI

The **Air Quality Index (AQI)** reports daily air quality and tells us how clean or polluted the air is, along with possible health effects. It measures major pollutants like ozone, particulate matter, carbon monoxide, sulphur dioxide and nitrogen dioxide. The EPA has set standards to protect health. Air pollution is a major threat causing respiratory disorders, heart diseases, and contributing to global warming.

WATER

After air, water is also essential for your life. You cannot live without water. It is possible to live without food for some days but it is very difficult to survive without water even for a day. About 70% of the human body is made up of water. Several life processes in living organisms take place in water.

Water is made up of two atoms of hydrogen and one atom of oxygen (H_2O).

Properties of Water

- Water is a colourless, tasteless and odourless liquid.
- It freezes at $0^{\circ}C$ and boils at $100^{\circ}C$.
- It can dissolve a large number of substances. So, it is known as a universal solvent.

Water Pollution

Water is said to be polluted if it contains impurities. The impure water is harmful for animals, plants and environment. These are of two types:

- Soluble impurities are salts and minerals. They dissolve in water.
- Insoluble impurities are stones, sand, mud, etc. They do not dissolve in water.

Following are the causes of water pollution:



- Disposal of waste materials from homes and factories into water bodies.
- Washing of chemical fertilizers and pesticides from the fields into the water bodies by rain or excess irrigation.

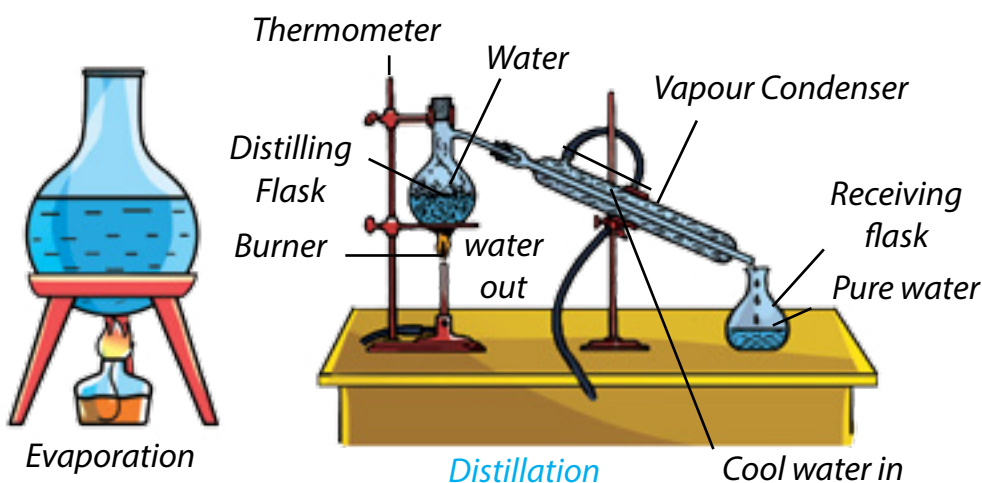
PURIFICATION OF WATER

Polluted water may contain many impurities like lead, mercury, sand, germs, bacteria, harmful gases, etc. All harmful impurities and germs (disease-causing) should be removed from water before drinking it. You should use the following methods:

Removal of Soluble Impurities

There are two methods of separating soluble impurities:

- **Evaporation:** It's a process to remove soluble impurities from water. In this, you heat the water solution with a mix of soluble impurities like salt. After some time, the water evaporates with air and the impurities are left behind.
- **Distillation:** In evaporation, you get only impurities and water mixes into air in the form of vapour. But in distillation, you can get impurities as well as pure water. In this process, contaminated water is heated in the flask. After some time, water changes into steam and impurities are left behind in the flask. This steam passes through a condenser. Here, it cools down and converts into water. This water is pure and is collected in a separate flask. This water can be used in medicines, car batteries, and science labs as it is pure, without any germs and impurities.



Distillation is the best method of getting pure water.

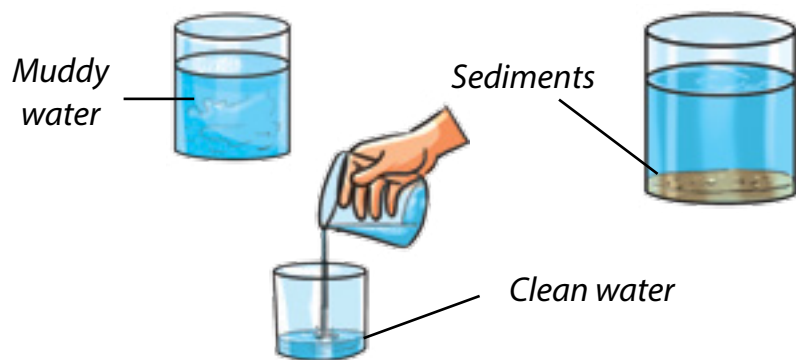
Removal of Insoluble Impurities

There are two methods to separating insoluble impurities:

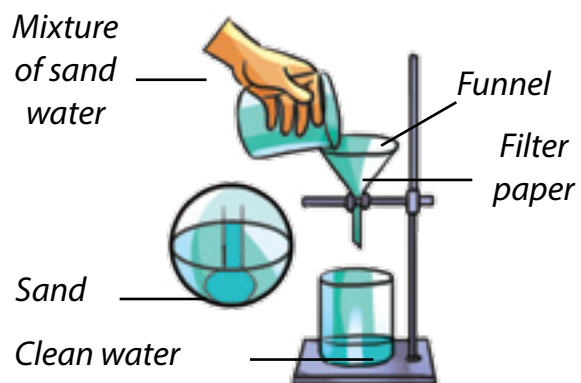
- **Sedimentation and Decantation:** Water with insoluble impurities is kept

undisturbed for some time. Impurities settle down and clean water comes up. This can be separated slowly by pouring out the clean water into another vessel as shown in the figure. Impurities are left behind.

- **Filtration:** Separation of insoluble impurities with the help of filter paper or filtering medium is known as **filtration**. For filtration, filter paper should be arranged in a cone shape. Fit this cone into a glass funnel. To make the filter paper stay in the glass funnel, sprinkle some water over it. Now, pour the mixture of sand and water in the funnel. The sand does not pass through the filter paper and remains behind the filter paper. This is called **residue**.



Sedimentation and Decantation



Filtration

Purifying Drinking Water

Water that is fit for drinking is known as potable water.

Following are some methods of purifying water:

- **Boiling:** You should drink boiled water. Boiling kills the germs present in water. It is the simplest way to purify water.
- **Chlorination:** You can use chlorine tablets in water to kill germs. It should be added in small quantities.
- **Water Purifier:** Water purifiers filter water and have ultraviolet rays tubes in them. UV rays kill the germs. This is an effective method.



Water purifier



Did You Know

By adding potassium permanganate crystals to wells to kill germs, we can purify water and make it fit for drinking.



Quick Reflection

Write two examples for each of the following.

1. Methods to remove insoluble impurities _____
2. Methods to remove soluble impurities _____
3. Methods to purify drinking water _____

Quick Recap

- * Air is essential for the existence of life.
- * The Earth is surrounded by a blanket of air called the atmosphere.
- * Air has weight and occupies space.
- * Air exerts pressure.
- * Air is a mixture of many gases.
- * Water is also very important for every living being without which you cannot survive.
- * Impurities of water must be removed before use.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. Air is a mixture of:
a) Oxygen and nitrogen ☐ b) Many gases ☐
c) Both a) & b) ☐
2. Which gas do humans breathe?
a) Carbon dioxide ☐ b) Oxygen ☐
c) Nitrogen ☐
3. How can you remove soluble impurities?
a) By evaporation ☐ b) By distillation ☐
c) Both a) & b) ☐
4. The process of removing impurities from water by using filter paper is called:
a) Distillation ☐ b) Filtration ☐
c) Boiling ☐

B. Fill in the blanks with the appropriate words from the box.

atmospheric oxygen ultraviolet Water temperature

1. The ozone layer in the atmosphere filters harmful _____ rays.
2. Atmosphere maintains the _____ on the Earth.
3. You breathe in _____ gas of the atmosphere to burn your food.
4. _____ is the universal solvent.
5. The pressure exerted by air on the Earth is known as _____ pressure.

C. Write T for true and F for false statements.

1. The main component of air is oxygen.
2. You should wash fruits and vegetables in plenty of water.
3. River water is potable water.
4. The purest form of water is distilled water.
5. Germs in the water are useful.
6. Acid rains are harmful to the plants and buildings.

☐
☐
☐
☐
☐
☐

D. Answer the following questions.

1. Name the four gases present in the air.
2. What do you mean by atmospheric pressure? List its uses.
3. What makes water impure? What are its harmful effects?
4. What do you mean by distillation? Explain the process.
5. How is acid rain formed?
6. How do electrical purifiers help to purify water?

CAPABILITY ANALYSIS

- A.** P&G sets a new goal to deliver 25 billion litres of clean drinking water to families in need worldwide. It is working with many partners in an effort to provide clean water to improve health and reduce illnesses caused by contaminated water. P&G is now accelerating its



efforts to help provide clean drinking water to even more people by delivering 25 billion litres—more than 100 billion glasses of water worldwide by 2025.

Environment and Health

Surf the Internet to learn more about the accomplishment of the goal, then prepare a Power Point presentation about it.

Digital Resources

B. To study the formation of solute, solvent and solution.

Experiential Learning

Things required: Four glasses, water, spoon, salt, sugar, sand and chalk powder.

Method: Take four glasses with equal amounts of water.

Add salt to one glass, sugar to second, sand to third, and chalk powder to fourth.

Stir them with a spoon.

You will notice that sugar and salt disappear in water whereas sand and chalk powder can still be seen at the bottom of the glasses.

Results: Here, salt and sugar are known as solutes, water is solvent and the resultant liquid is known as a solution.

C. Discuss the methods of water purification that you and your classmate

Communication skills

D. Make a rainstorm project and observe the precipitation.

Multidisciplinary Learning

- Fill the jar with water.
- Squirt foam shaving cream over the top so as it completely covers the surface.
- Fill a cup with water and add plenty of blue food colouring.
- Using a dropper, dribble blue water on cream and watch the rainstorm form.

E. Make a list of the various rivers in India. Classify them as rain-fed or s

Knowledge of India

F. Make a poster with a suitable picture and slogan, urging people to prevent water pollution.

G. Our most valuable resource is water. Each and every living thing needs water. Make a poster about water conservation.

Art Integration

HOTS

Empty bottles of mineral water or some other plastic bottles should be crushed before throwing them into the dustbin. Why?

Life Skill

Our Sky

Objectives



- The Moon
- Satellites

- Eclipses
- Chandrayaan-3 Mission

If you look at the sky during the day and night, you can see many objects and a few events taking place in the sky which make up our universe. The solar system is made up of eight planets. Our Earth is one of the planets. The Sun is at the centre of our solar system. In the night sky, you can see moon and thousands of twinkling stars.



Learn the different phases of the moon by using Oreo cookies! In this edible astronomy activity, you will recreate the different phases of the moon with yummy cookie sandwiches.

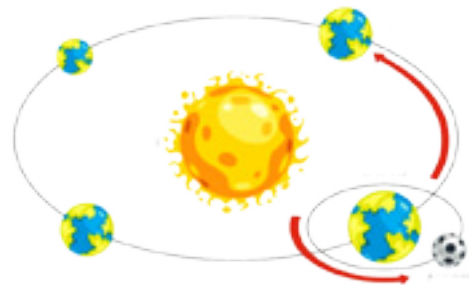
Hands-On Learning

1. Separate 8 Oreo cookies by slowly twisting the top and bottom cookies.
2. Use the knife or popsicle stick, carefully scrape off a portion of the frosting to recreate each moon phase.
3. Place the Oreo cookies on a paper plate in the correct order. Start off with the new moon and go counter clockwise from there.



THE MOON

As the Earth goes around the sun, the moon goes around the Earth. The moon is our Earth's closest neighbour in space. The moon is the only one natural **satellite** of our Earth. The distance of the moon from the Earth is 3,84,400 km. There is no life on the moon as it has no atmosphere.



Surface of the Moon

The moon is a ball of grey rocks that are covered with dust. Its surface is made up of mountains and plains. Huge saucer shaped holes are scattered over the surface of the moon. They are called **craters**.



Satellite : bodies that go around planets

The sky on Earth is blue because the atmosphere scatters the light from the Sun. There is no air to scatter light on the moon, so even in the daytime, the sky is pitch black and filled with stars. Due to the absence of atmosphere, the surface of the moon becomes extremely hot during the daytime and extremely cold during the night. With no air, there is no sound or wind to stir the rocky dust. Without air and water, no life can exist on the moon. So, the moon is still a barren and lifeless place.



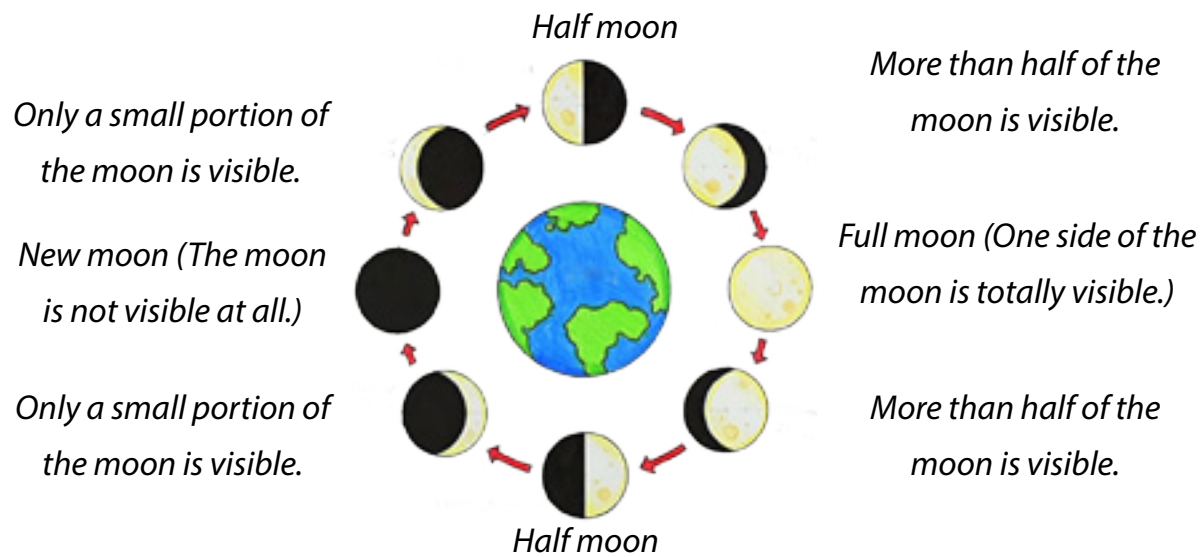
Phases of Moon

Every night, the moon appears to change its shape. The day-to-day changes in the shape of the moon, as seen from the Earth, are called **phases of the moon**.

The different phases of the moon are as follows:

- **New moon:** When the moon is between the Earth and the sun, you cannot see the moon at all. This is called **new moon**. It is dark on the New Moon day.
- **Crescent moon:** When you see a small sun-lit portion of the moon, it is called **crescent moon**.
- **Half moon:** When you see half of the moon, it is called a **half moon**.
- **Gibbous moon:** When you see over half the moon, it is called the **gibbous moon**.

- **Full moon:** When you see a complete moon, it is called a **full moon**.



? Did You Know

Neil Armstrong, Edwin Aldrin and Michael Collins went on the moon in a spacecraft named APOLLO 11, on 21st July, 1969. Neil Armstrong was the first person to land on the moon.



The Moon and the Tides

- The moon has weak gravity which has a noticeable effect on the Earth. It pulls the water from the seas and oceans of the Earth, causing **tides**. Since the moon is closer to the Earth than the Sun, the moon's pull on water is stronger than that of the sun. The water of the seas facing the moon is pulled upwards raising the sea level. This is known as a **high tide**. On the opposite side of the Earth, another high tide occurs due to the movement of the Earth.
- The water forms low tides as its level falls. In the areas between the two high tides, sea during low tide and come back to the coast during high tide.

? Did You Know

The moon's gravity is 1/6th of the Earth.



tide : a regular rise and fall of water in the sea.

ECLIPSES

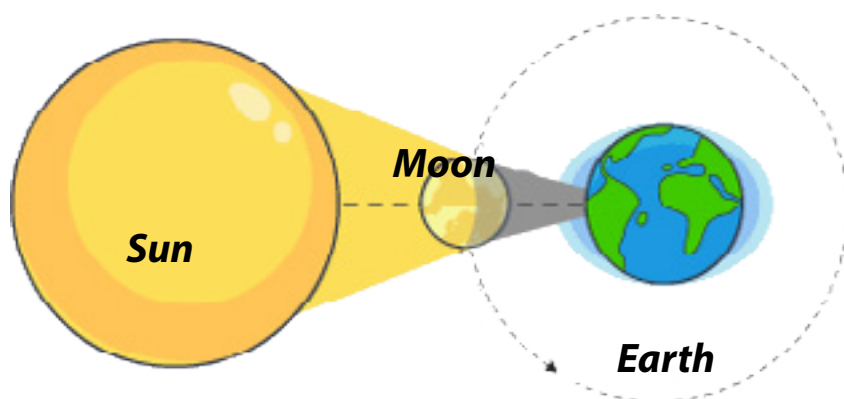
In presence of light, you see the shadow of an object. Some objects cast dark shadows. Some cast light shadows. Objects through which light cannot pass are called **opaque objects**. They form dark or clear shadows, e.g., wood, human bodies, etc.

An **eclipse** is also a shadow but formed in space between the sun or moon and the Earth. The Earth and the moon keep changing their positions in the space. Sometimes, the Earth comes in between the moon and the sun and sometimes the moon comes in between the Earth and the sun.

The object that is in the middle of the line will cast a shadow that partially or completely hides the other from the sun. This is called an **eclipse**. An eclipse can be partial or total.

Eclipse is of two kinds: • Solar Eclipse • Lunar Eclipse

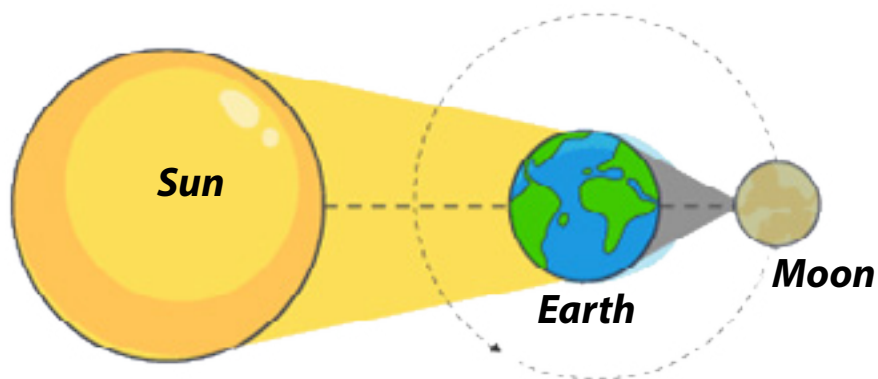
Solar Eclipse : In a solar eclipse, the moon comes between the sun and the Earth. Thus, it obstructs the sun's light from falling on the Earth. The moon casts its shadow on the Earth. You see only a part of the sun.



When you cannot see the sun, you call it a **total solar eclipse**. If you can see a part of the sun, you call it a **partial solar eclipse**.

Lunar Eclipse: A lunar eclipse

takes place when the Earth comes in between the sun and the moon. The Earth casts a shadow on the moon and it becomes invisible. This is a lunar eclipse. When the Earth's



shadow covers the moon completely, you have a total lunar eclipse. If it covers the moon partially, you have a partial lunar eclipse.

SATELLITES

Satellites are both man-made and natural. The moon is the only natural satellite of the Earth that revolves around it.



Quick Reflection

1. What is the natural satellite of the Earth?
2. How is shadow formed?

Artificial satellites are man-made machines sent into space by rockets to revolve around the Earth just like the moon. The first satellite Sputnik I was launched by Russian scientists. Since then, many satellites have been launched. Aryabhata, Bhaskara, INSAT IA, INSAT IB, Apple, and Rohini are different satellites sent by India into space.



INSAT IA



Apple

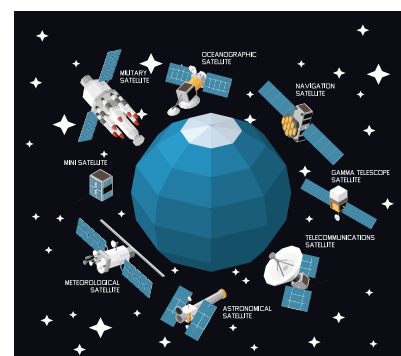


Rohini

Uses of Satellites

Initially, artificial or man-made satellites were created for scientific study and to provide us with useful data regarding the Earth's atmosphere.

There are different types of satellites, each designed for a specific purpose. Satellites that send messages from one country to another are called **telecommunication satellites**. Such satellites send signals for TV and radio programmes and telephone calls. **Navigation satellites** help ships and air crafts find their way. **Astronomical satellites** carry telescopes into space. Astronomers use such satellites to take special measurements of our solar system, stars and galaxies. **Meteorological satellites** gather data concerning the earth's atmosphere and surface in order to aid meteorologists in understanding weather patterns and producing weather forecasts. **Military satellites** are used for common missions such as intelligence gathering, navigation and military communications.



Different types of satellites

CHANDRAYAAN-3 MISSION

Spacemen have to do different jobs. Some do experiments, some work inside the shuttle, some do repair outside the spacecraft. Others even repair satellites. The astronauts return to earth on space shuttles after their mission is over. Rockets can only be used once whereas shuttles can be used repeatedly.

The Indian Space Research Organization (ISRO) launched Chandrayaan-3 to explore the moon with a soft landing. Building upon Chandrayaan-1 and Chandrayaan-2, this mission pushes the boundaries of space discovery. It took off from Satish

Dhawan Space Centre on July 14, 2023, aboard a GSLV-MK III heavy-lift rocket.

Chandrayaan-3 carries instruments to study lunar soil properties, seismic activity, surface environment, and elemental composition near the landing site. The spacecraft entered lunar orbit on 5 August, and the lander Vikram touched down near the south pole on 23 August 2023, making India the fourth nation to land on the moon and the first near its south pole.



Astronaut in spacesuit



Did You Know

On Earth, telescopes are employed to provide a clear picture of space. There are now floating Hubble telescopes in space which provide a clearer perspective and more information.



Chandrayaan-3 on the moon

Quick Recap

- * The sun is at the centre of the solar system.
- * The moon is the only one natural satellite of our Earth.
- * Shadows are formed when something comes in the way of light.
- * An eclipse is a kind of shadow.
- * A solar eclipse occurs when the moon comes between the sun and the Earth.
- * A lunar eclipse occurs when the Earth comes between the sun and the moon.
- * The Indian Space Research Organization (ISRO) has launched ambitious lunar mission Chandrayaan-3 to explore the Moon on July 14, 2023.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

- Which of the following is at the centre of the solar system?
a) Earth ☐ b) Sun ☐
c) Moon ☐
- Huge saucer-shaped holes scattered over the surface of the moon are called:
a) glaciers ☐ b) holes ☐
c) craters ☐
- When you see more than half of the moon, it is called:
a) half moon ☐ b) gibbous moon ☐
c) new moon ☐
- Which of the following is not a satellite?
a) INSAT 1A ☐ b) INSAT 1B ☐
c) NASA ☐

B. Write T for true and F for false statements.

- Moon is the man-made satellite of the Earth. ☐
- There is no life on the moon as it has no atmosphere. ☐
- Shadows are formed when light is obstructed. ☐
- When the shadow of the Earth falls on the moon, a solar eclipse occurs. ☐
- The sky is blue because of the presence of water on Earth. ☐
- An eclipse can be partial or total. ☐

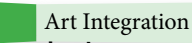
C. Give reasons.

- You cannot see through an opaque object.
- Oceans and seas experience tide.
- Shadow do not form when light hits a transparent object.

D Answer the following questions.

1. Describe some features of the moon.
2. What do you mean by tides? How do tides happen?
3. How are tides useful?
4. What is an eclipse? How does solar eclipse happen?
5. How are artificial satellites useful? Name some artificial satellites.

CAPABILITY ANALYSIS

- A.** An eclipse is a kind of shadow. There are two types of eclipses-lunar eclipses and solar eclipses. Draw the labelled diagrams of the lunar and solar eclipse on chart paper.  Art Integration
- B.** “Satellites are helpful to us in many ways.” Organize a discussion on this topic after doing a search on Google about satellites.  Communication Skills

C. SpaceX satellite Internet service-Starlink

Starlink is a satellite Internet system that is operated by Elon Musk’s rocket company, SpaceX. It started launching Starlink satellites in 2019. As of May 2023, Starlink consists of over 4,000 mass-produced small satellites in low Earth orbit (LEO).



Investigate Elon Musk’s building a satellite constellation online and write a report. Digital Resources

- D.** The Indian Space Research Organization (ISRO) has launched ambitious moon mission Chandrayaan-3. The Chandrayaan-3 mission carries scientific instruments to study the thermo-physical properties of the lunar regolith, lunar seismicity, lunar surface plasma environment and elemental composition in the vicinity of the landing site.



Surf the Internet to learn about the Chandrayaan-3 mission and its objectives.

Then prepare a Word document for the same.  Digital Resources

E. Let's make a Sparkly Galaxy Playdough!

Creativity

Use separate colours to recreate this playdough. You can also use food colouring and cream of tartar. Shades of pinks, purples and blue blend together for a galaxy look. Once it is kneaded, add your glitter and mix until it's fully incorporated. Use plastic knives and cookie cutters. Enjoy the playdough!

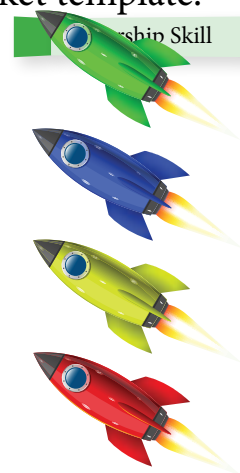


F. Let's make straw rockets and see how far they could go out at different angles!

Materials required: Straws, tape, markers, crayons, scissors and a rocket template.

Method: Make a group of four and colour your rocket in your style. Make sure your straw is wider than the other straw you'll be using for launching. Cut the straw to fit the length of the rocket and tape one end shut so it's completely sealed. Attach it to your rocket with tape. You're ready to launch! Give your straw a big puff of air and watch it take off!

Compare and contrast the different angles and varied fins used to record the results. Infer how the straw affects the distance the rocket travels or how adding fins or folding the rocket in different ways can affect its flight. Compete and record who leads among the four?



G. Gather data regarding NASA's Mars mission. Do some Internet research on it, and then create a Word document for it.

Digital Resources

H. The Evolution of the Spacesuit

Engineers working to design a new spacesuit for future missions have come up with a suit system that consists of 2 basic configurations that will be used for different tasks.

The orange suit is Configuration 1, which will be worn during launch and landing. It will also be used if a space walk must be performed in microgravity. Configuration 2, the white suit, would be used during moonwalks for lunar exploration. Since Configuration 1 will be used in and around the vehicle only, it does not need the life support backpack that Configuration 2 uses – instead it will connect to the vehicle by umbilical. The MK III is an advanced demonstration suit being used to develop elements for future suits.



Learn more about the engineers, scientists and astronauts who are creating prototype spacesuits for simulating scientific activity as part of NASA's demonstration of ideas for living and working on the lunar surface.

HOTS

Multidisciplinary Learning

If you can jump to a height of a metre on Earth, you can jump 6 metres high on the moon.
How is it possible?

Scientific Temper

Light, Shadow, and Sound

Objectives



- Light
- Shadow
- Types of Objects
- Sound

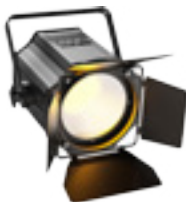
We cannot see anything without light. In complete darkness, objects remain invisible to us. Light is essential for vision, and the sun is the primary natural source of light on Earth.

The brightness of a light source depends on the amount of light it produces. The more light it emits, Tick (✓) the brighter light source in each pair.

Critical Thinking


☐
☐

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☐


LIGHT

The sun is the main natural source of light during the day. As the sun sets, darkness spreads across the surroundings. The moon appears in the night sky, but it does not generate its own light. Instead, it reflects the sunlight.

Luminous and Non-Luminous Objects

Man-made sources of light include bulbs, lamps, diyas, tube lights, and candles. Objects that emit light are called **luminous objects**, such as the sun, tube lights, lamps, and candles.



The Objects that do not produce light are known as **non-luminous objects**. Examples include stone, wood, pens, paper, cloth, bottles, and bags.



TYPES OF OBJECTS

We encounter different types of objects in our daily lives. Some allow light to pass through them, while others do not. Based on how much light they permit, objects are categorized into three types:

1. Transparent Objects

Objects that allow light to pass through completely are called **transparent objects**. Examples: glass, clean water.

2. Translucent Objects

Objects that permit some light to pass through are called **translucent objects**.
Examples: an oiled sheet of paper, frosted glass, dirty water.

3. Opaque Objects

Objects that do not allow any light to pass through them are called **opaque objects**.
Examples: wood, mirrors.

? Did You Know

Transparent objects do not cast shadows because they do not block light.

SHADOW

Shadow is a dark outline or image formed when an opaque or translucent object blocks light. Shadows always appear in the opposite direction of the light source. Regardless of an object's colour, its shadow is always black.



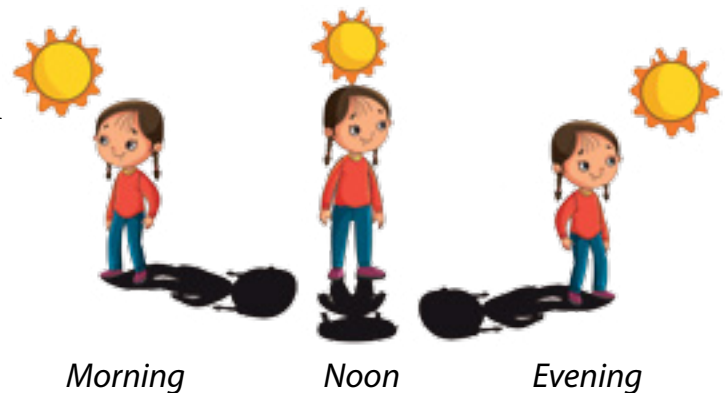
Slanting : Positioned at an angle or slope.
Honk : A loud sound produced by a vehicle's horn.

Size of a Shadow

The size of a shadow varies depending on the position of the light source.

Slanting rays (when the sun rises or sets) create long shadows.

Direct rays (at noon) create short shadows.



Morning

Noon

Evening

SOUND

Sound is what we hear and helps us communicate. We experience different sounds daily, such as **honking** vehicles, mobile ringtones, chirping birds, and barking dogs.

? Did You Know

Solar eclipse and lunar eclipse are the results of shadows.



Pleasant sound

Some sounds are pleasant and soothing, while others are loud and irritating. Unpleasant, disturbing sounds are known as noise. To maintain a peaceful environment, we should keep television and music systems at a low volume.



Unpleasant sound

Quick Recap

- * The sun is the primary source of light on Earth.
- * Objects that emit light are called luminous objects.
- * Based on the amount of light they allow through, objects are classified as transparent, translucent, or opaque.
- * Objects that do not produce light are non-luminous objects.
- * A shadow is a dark image of an object that forms when light is blocked.
- * Sounds can be pleasant or unpleasant (noise).

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. What is the main source of light on Earth?
a) Sun ☐ b) Torch ☐
c) Candle ☐
2. Which of the following is a luminous object?
a) Stone ☐ b) Cloth ☐
c) Candle ☐
3. Which of the following is a translucent material?
a) Wood ☐ b) Mirror ☐
c) Coloured glass ☐
4. What is the dark image of an object called?
a) Force ☐ b) Shadow ☐
c) Sound ☐
5. Which of these does not form a shadow?
a) Clear water ☐ b) Wood ☐
c) Mirror ☐

B. Fill in the blanks with the appropriate words from the box.

sound light opposite non-luminous

1. Objects that do not **emit** light are called _____.

2. A shadow is formed on the _____ side of an object.
3. A shadow is formed when something blocks the way of _____.
4. When we speak to someone, we produce _____.

C. Write T for true and F for false statements.

1. A sheet of paper rubbed with oil is transparent.
2. Candles, torches, and bulbs are non-luminous objects.
3. A shadow is a colourful image of an object.
4. Shadows are shortest at noon.
5. Pleasant sounds are called noise.

☐
☐
☐
☐
☐

D. Answer the following questions.

1. What are luminous and non-luminous objects?
2. What are the three types of objects based on how much light they allow through?
3. How is a shadow formed?
4. What is noise?

CAPABILITY ANALYSIS

A. Tick the objects that can reflect light and form a reflection.

Critical Thinking

☐☐☐☐☐☐

B. Draw pictures of shadows at different times of the day and paste them in your scrapbook.

Experiential Learning

C. Paste pictures of transparent, translucent, and opaque objects in your scrapbook.

Art Integration

D. Tick (✓) the activities that produce noise.

Critical Thinking

1. A girl singing a song.
2. The voice of street vendors in the market.
3. Honking vehicles on the road.
4. Talking to each other in class.

☐
☐
☐
☐

E. Identify the animals by looking at their shadows and write their names.

1.



2.



3.



4.



F. Fun Activity: Making Rainbow Candles for Diwali!

A rainbow is a colourful arc that appears in the sky when sunlight is refracted through raindrops after rainfall.

Vocational Skill

Let's make Rainbow Candles!

- Gather materials – crayons, wax, moulds.
- Melt the wax using a double boiler.
- Layer the colours in the mould.
- Let it cool – your colourful rainbow candle is ready!



HOTS

Why don't transparent objects form shadows?

Critical Thinking

Natural Disasters

Objectives



- *Understanding Natural Disasters*
- *Disaster Management*

We frequently hear in the news about major destruction caused by heavy rainfall, earthquakes, droughts, and other natural events. These are natural occurrences that happen periodically.

Match the following natural disasters pictures with their names:

Hand-On Learning

1.



a) Landslide

b) Tornado

2.



3.



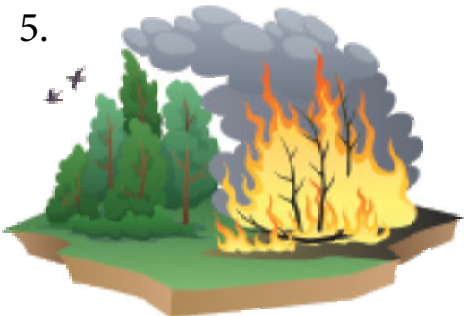
c) Volcano

d) Flood

4.



5.



e) Wildfire

f) Drought

6.



These natural occurrences sometimes result in significant damage to life and property.

NATURAL DISASTERS

A natural disaster is a severe event that occurs due to natural phenomena, leading to significant destruction and loss of life. Examples include earthquakes, floods, droughts, cyclones, and tsunamis.

Floods

Floods occur when an excessive amount of water overflows onto land,

often causing immense destruction. The strong water currents can damage buildings and trees. Heavy rainfall or overflowing rivers are common causes of **floods**. Water logging from floods can persist for several days, making the environment unhealthy.



Did You Know

Floods can be controlled by planting more trees and constructing check dams.



Flood

Volcanoes

The Earth's core contains molten rock. A volcano is a crack in the Earth's crust that allows magma, gases, and ash to escape. When this molten rock reaches the surface, it is known as **lava**. Volcanic mountains typically have a conical shape. The intense pressure inside the Earth forces magma upward, erupting through a **vent** in the volcano.



A volcano on the surface of the earth.

Types of Volcanoes:

Active Volcanoes: These erupt frequently or have erupted in the recent past. About 500 active volcanoes exist today, primarily around the Pacific Ocean in an area known as the **Pacific Ring of Fire**. Examples include Mount Vesuvius and Mount Fuji. Barren Island in the Andaman and Nicobar Islands is the only active volcano in the Indian subcontinent.

Dormant Volcanoes: These volcanoes are currently inactive but may erupt in the future. Examples include Narcondam in the Andaman Islands and Mount Kilimanjaro in Tanzania.



Mount Fuji

Extinct Volcanoes: These no longer erupt due to a lack of lava supply. Examples include the Emperor Sea mount Chain in the Pacific Ocean and Zuidwal in the Netherlands.

Earthquakes

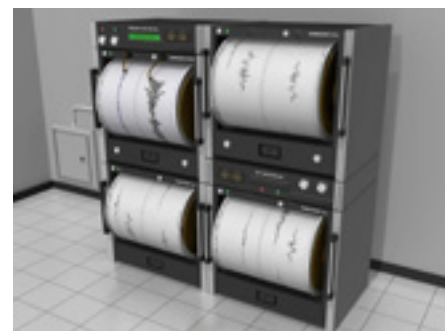
An **earthquake** occurs when the Earth's surface shakes due to shock waves travelling through solid rock. The point inside the Earth where an earthquake originates is called the **focus**, while the area directly above it on the surface is known as the **epicenter**. The greatest destruction occurs in the epicenter area. Following a major earthquake, smaller tremors known as **aftershocks** can occur for several days. For example, on January 26, 2001, Gujarat (particularly Bhuj) experienced a devastating earthquake.



Earthquake

Measuring Earthquakes

The **seismograph** is an instrument used to measure earthquakes. The intensity is recorded on the **Richter scale**. Earthquakes measuring 5 or above on the Richter scale are considered highly destructive.



Seismograph

EFFECTS OF EARTHQUAKES

- Severe earthquakes cause damage to buildings, bridges, roads, and forests.
- Large cracks form on the Earth's surface.
- Power lines snap, and fires may break out in affected regions.

Tidal Waves and Tsunamis

Tidal waves refer to the periodic rise and fall of sea levels. Tsunamis are massive ocean waves triggered by undersea earthquakes, landslides, or volcanic eruptions. The water speed in a tsunami can reach 800 km per hour, causing widespread destruction when it reaches land. A devastating tsunami occurred on December 26, 2004, affecting countries like India, Sri Lanka, Bangladesh, and Thailand.



Tsunamis

Droughts

Drought occurs when an area receives very little rainfall, causing rivers, lakes, and underground water to dry up. This leads to cracks in the Earth's surface and failed crops, resulting in food shortages and famine.



Droughts

Cyclones

A **cyclone** is a powerful storm that rotates in a spiral motion due to storms over the sea. These storms move quickly toward coastal regions, causing extensive damage to infrastructure, power supply, and communities.



Cyclones

DISASTER MANAGEMENT

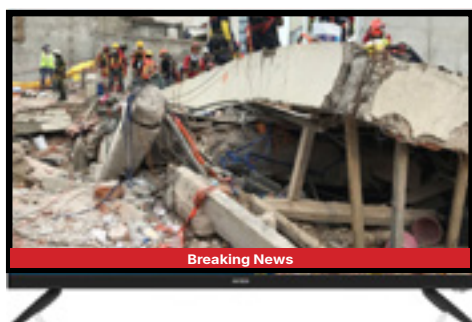
Natural disasters hinder a country's progress. To minimize damage, disaster preparedness and response are essential. During such crises, different agencies and professionals play crucial roles:

- Doctors provide medical assistance to the injured.
- Police assist in rescue efforts.
- Fire fighters help put out fires.
- Armed forces distribute food, clothing, and supplies in disaster-hit areas.
- Government, businesses, schools, and colleges contribute by donating money and resources.
- Weather forecasts help predict natural disasters like floods and cyclones in advance.
- Media keeps people informed and encourages aid efforts.



Air Force during natural disasters

Several organizations offer relief during disasters, including:



- Red Cross Society
- World Health Organization (WHO)
- United Nations Children's Emergency Fund (UNICEF)

Quick Recap

- * A natural disaster is a catastrophic event caused by nature.
- * Volcanoes are cracks in the Earth's crust through which magma escapes.
- * Earthquakes are caused by shock waves beneath the Earth's surface.
- * A seismograph measures earthquake intensity.
- * Tsunamis, also known as tidal waves, are massive ocean waves caused by underwater disturbances.
- * Droughts occur due to insufficient rainfall.
- * Cyclones are severe storms with spiralling winds.
- * Many organizations work to provide aid and support during disasters.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

- Which of the following is a natural disaster?

a) Earthquake ☐	b) Fire ☐
c) Both a) & b) ☐	
- What causes an earthquake?

a) Shock waves ☐	b) Aftershocks ☐
c) Floods ☐	
- The molten rock that flows on the Earth's surface is called:

a) Magma ☐	b) Lava ☐
c) Volcano ☐	
- Zuidwal in the Netherlands is:

a) An extinct volcano ☐	b) A dormant volcano ☐
c) An active volcano ☐	
- Which organization supports disaster relief efforts?

a) UNCE ☐	b) UNICEF ☐
c) UNIQUE ☐	

B. Give one-word answers.

1. Giant waves in the sea caused by an earthquake: _____
2. A volcano that erupts frequently: _____
3. Small tremors following a major earthquake: _____
4. A disaster caused by undersea earthquakes: _____
5. The point above the earthquake's focus: _____

C. Match the following.

Column A

1. Molten rock inside
2. Molten rock outside
3. Tsunami
4. Earthquake

Column B

- (a) Richter Scale
- (b) Lava
- (c) Magma
- (d) Tidal waves

D. Fill in the blanks with the appropriate words from the box.

weather forecast

starvation

rainfall

focus

1. Floods occur due to excessive _____.
2. The underground location where an earthquake originates is known as the _____.
3. _____ is a consequence of droughts.
4. The _____ provides up-to-date information regarding natural disasters.

E. Answer the following questions.

1. What is a natural disaster?
2. What are the three types of volcanoes?
3. How do earthquakes occur?
4. What are aftershocks?
5. What is a seismograph?
6. What is a tsunami?
7. What is a drought? Explain.
8. List two organizations that assist people during disasters.

CAPABILITY ANALYSIS

- A.** Gather newspaper clippings related to ‘Natural Disasters’ and paste them onto a chart paper.

Art Integration

(This task involves collecting real-life news reports about natural disasters.)

- B.** Divide the class into groups and discuss: “How can we assist those affected by natural disasters?”

Communication Skill

(Students should share ideas and strategies for providing disaster relief.)

- C.** Establish a Disaster Management Group for classes I to V.

Leadership Skill

- Include enthusiastic students, teachers, and parents.
- Form a committee for disaster management.
- Organize training on disaster preparedness.
- Conduct periodic meetings and discussions.

- D. Create a rainbow baking soda volcano!**

Hand-On Learning

(This experiment demonstrates a reaction between vinegar and baking soda, which produces carbon dioxide gas, leading to a colourful volcanic eruption.)



- E.** Research safety measures for earthquakes and design an informative chart.
(The chart should highlight essential safety steps such as “Drop, Cover, and Hold,” keeping away from windows, and preparing an emergency survival kit.)

Art Integration

HOTS

Why does drought often lead to famine?

Critical Thinking

Our Environment

Objectives



- Pollution
- Types of Pollution
- Ways to Control Pollution

The **environment** includes everything around us that influences the lives of plants and animals. Unfortunately, the quality of air and water has been deteriorating due to various factors.

Human activities play a significant role in affecting the environment.

Hand-On Learning

One important concept is Carbon Footprint, which refers to the amount of carbon emissions generated by an individual's daily activities. These emissions include carbon dioxide and other gases that trap heat in the atmosphere, contributing to global warming.

Environment and Health

There are several ways to reduce your carbon footprint and help protect the environment. Can you think of some actions that can help lower carbon emissions?



What is the distinction between an ecological footprint and a carbon footprint?

Multidisciplinary Learning

Do dietary choices influence your carbon footprint?

Critical Thinking

Does the use of electricity at home or school increase carbon emissions?

Critical Thinking

UNDERSTANDING THE ENVIRONMENT

The environment is made up of both living and non-living elements. Living components include plants, animals, and humans, while non-living elements consist of soil, water, air, and sunlight. All these components interact with one another, forming a balanced ecosystem.

POLLUTION

Pollution occurs when harmful substances contaminate air, water, or land. These harmful substances are known as **pollutants**.

TYPES OF POLLUTION

There are four main types of pollution:

Air Pollution

Air pollution results from the introduction of toxic substances into the air due to human activities. Major causes include industrial emissions, vehicle exhaust, burning of waste, smoking, forest fires, and household fires.

It can lead to severe health issues like asthma, lung cancer, and other respiratory diseases.



Water Pollution

Water pollution occurs when harmful substances are introduced into water bodies due to human activities. The primary causes include:

- Industrial wastewater released into rivers, lakes, or oceans.
- Pesticides, fertilizers, and herbicides from agriculture contaminating water sources.
- Domestic activities like washing clothes or bathing animals in water bodies.
- Sewage and untreated household waste directly dumped into water bodies.



Land Pollution

Land pollution refers to the **degradation** of land due to human actions. It is mainly caused by:



degrade : the process in which quality of something is destroyed

- Improper agricultural practices
- Industrial waste dumping
- Garbage dumped in public areas
- Deforestation



Noise Pollution

Excessive and unpleasant noise is termed **noise pollution**. It originates from traffic, loudspeakers, factory operations, and other loud activities. Noise pollution can cause stress, irritation, and even hearing loss.



CONTROLLING POLLUTION

Pollution can be minimized through the following measures:

- Controlling population growth.
- Relocating factories away from residential areas.
- Installing tall chimneys in factories to release gases safely.
- Disposing of waste properly in designated areas.
- Avoiding the open burning of garbage.
- Replacing plastic bags with cloth or paper alternatives.
- Planting more trees to increase oxygen levels.
- Using water responsibly.
- Encouraging vehicles to use CNG (Compressed Natural Gas) to reduce emissions.



Quick Recap

- * The environment consists of both living and non-living things.
- * Human activities impact the environment significantly.
- * Pollutants are harmful substances that contaminate air, water, and land.
- * The four primary types of pollution are air, water, land, and noise pollution.
- * Pollution can be reduced through responsible actions.

ABSTRACT ANALYSIS

A. Tick (✓) the correct option.

1. Activities that contribute to air pollution:

- a) Industrial smoke ☐ b) Vehicle exhaust ☐
c) Both (a) & (b) ☐

2. Burning garbage leads to:

- a) Land pollution ☐ b) Noise pollution ☐
c) Air pollution ☐

3. To reduce pollution, vehicles should use:

- a) Petrol ☐ b) Diesel ☐
c) CNG ☐

4. To minimize pollution, we should use:

- a) Plastic bags ☐ b) Jute bags ☐
c) Both (a) & (b) ☐

5. _____ can cause hearing loss.

- a) Air pollution ☐ b) Noise pollution ☐
c) Water pollution ☐

B. Fill in the blanks with suitable words.

1. Two natural resources: _____
2. Materials that pollute the environment: _____

3. Two diseases caused by air pollution: _____
4. Chemicals released from industries: _____
5. The process of soil erosion: _____

C. Write T for true and F for false statements.

1. Planting more trees helps the environment. ☐
2. Using cloth bags reduces pollution. ☐
3. Air pollution does not cause lung cancer. ☐

D. Answer the following questions

1. What is pollution?
2. What are pollutants?
3. What are the four major types of pollution?
4. How does air pollution affect human health?
5. What are the main causes of water pollution?
6. What is soil pollution, and what causes it?
7. List three ways to control pollution.

CAPABILITY ANALYSIS

A. Create a Scrapbook

Art Integration

Find and paste pictures of different pollution-causing activities under the appropriate categories.

B. Design a Poster

Creativity

Create an eye-catching poster with the theme “Save Mother Earth”.

C. Organize a Discussion

Environment and Health

Conduct a classroom debate on “Pollution and Its Harmful Effects”.

D. Research the Kyoto Protocol

Digital Resources

Find out about the Kyoto Protocol, an international agreement to reduce greenhouse gas emissions. Prepare a Power Point presentation to educate your classmates.



E. Alternative Fuels for Vehicles

Digital Resources

Since fuel-powered vehicles significantly contribute to pollution and global warming, we must explore alternatives. Some alternatives to fossil fuels include:

- Natural gas
- Electric vehicles
- Hydrogen fuel cells
- Bio-diesel

Conduct research on alternative fuels and compile your findings into a Word document.



F. Understanding Air Purifiers

The COVID-19 pandemic has increased awareness of indoor air quality. Scientific research suggests that Aura Air removes 99.9% of viruses, bacteria, and pollutants, improving indoor air quality.



- Explore different air purifier models and note their specifications and technologies.
- Do air purifiers effectively remove odours and pollutants?
- Are there any limitations to air purifiers?

Scientific Temper

Environment and Health

Critical Thinking

HOTS

How does planting trees help in reducing pollution?

Scientific Temper

Test Paper 1

(Based on Chapter 1-6)

A. Tick (✓) the correct option.

1. How are cotton seeds primarily dispersed?
a) By air ☐ b) By water ☐
c) By explosion ☐
2. Which condition requires iodine in salt to prevent it?
a) Anaemia ☐ b) Goitre ☐
c) Beri-beri ☐
3. Which of these is a good conductor of electricity?
a) Mud ☐ b) Sand ☐
c) Water ☐
4. Which metamorphic rock originates from granite?
a) Gneiss ☐ b) Basalt ☐
c) Mica ☐
5. The specific place and surroundings where an animal resides are known as its _____.
a) Home ☐ b) Habitat ☐
c) Shelter ☐
6. Which is the largest part of the brain?
a) Cerebellum ☐ b) Cerebrum ☐
c) Medulla ☐

B. Match the following.

Column A

1. Seed dispersed by animals
2. A Rabi crop
3. Seed dispersed through explosion
4. Young plant
5. Seed dispersed by water

Column B

- a) Seedling
- b) Coconut
- c) Poppy
- d) Pea
- e) Xanthium

C. Fill in the blanks with appropriate words from the box.

spiracles

framework

Bryophyllum

fossil

vitamin C

viruses

1. The _____ plant grows from leaf buds.
2. Scurvy results from a deficiency of _____.
3. The saliva of animals contains _____.
4. Coal is a type of _____ fuel.
5. Insects respire through _____.
6. The skeletal system serves as the _____ of the body.

D. Provide a one-word answer.

1. The two hind limbs of an animal _____
2. The animal known as the 'ship of the desert' _____
3. A flightless bird _____
4. A network of thin tubes connected to spiracles _____
5. An organism that breathes through both lungs and skin _____

E. Identify and name the various types of joints in the human body.

1.



2.



3.



4.



F. Answer the following questions.

1. Explain the process of germination.
2. What is PEM?
3. Define first-aid.
4. Describe the formation of petroleum and its derivatives.
5. What is migration, and why do animals migrate?
6. Describe the function of the spinal cord.
7. Differentiate between voluntary and involuntary muscles.

Test Paper 2

(Based on Chapter 7-13)

A. Tick (✓) the correct option.

1. The ultimate source of energy on the Earth is _____.
a) sun ☐ b) water ☐
c) wind ☐
2. It's a rod passing through a wheel.
a) Axle ☐ b) Wedge ☐
c) Lever ☐
3. How can you remove soluble impurities?
a) By evaporation ☐ b) By distillation ☐
c) Both a) & b) ☐
4. When you see more than half of the moon, it is called:
a) half moon ☐ b) gibbous moon ☐
c) new moon ☐
5. _____ is a translucent material.
a) Wood ☐ b) Mirror ☐
c) Coloured glass ☐
6. The molten rock that flows on the surface of the Earth is called:
a) magma ☐ b) lava ☐
c) volcano ☐

C. Write T for true and F for false statements.

1. Gravitational force attracts the iron objects. ☐
2. Washing machine is a complex machine. ☐
3. The purest form of water is distilled water. ☐
4. There is no life on the moon as it has no atmosphere. ☐
5. Shadow is longest at noon. ☐
6. Factories should be shifted to residential areas. ☐

B. Match the following.

Column A

1. 
2. 
3. 
4. 
5. 

Column B

- a) Wind energy
- b) Buoyant force
- c) Magnetic force
- d) Light energy
- e) Electrostatic force

D. Give two examples of each.

1. First-class lever: _____
2. Simple machines: _____
3. Wedge: _____
4. Inclined plane: _____
5. Complex machines: _____

E. Name the following.

1. Chemical disposed from factories and industries: _____
2. Materials that pollute the environment: _____
3. Two diseases caused due to air pollution: _____
4. A natural disaster caused by undersea earthquakes: _____
5. Wearing away of topsoil: _____

F. Answer the following questions.

1. List the four effects that force has on objects.
2. Give two uses of an inclined plane.
3. What do you mean by distillation? Explain the process.
4. What is an eclipse? How does a solar eclipse happen?
5. What is noise?
6. Name the three types of volcanoes.
7. Write any two measures to control pollution.